Geoeconomic Coercion: Which Target, Who Pays?

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Abstract

This paper studies the incidence of economic sanctions, across countries and within each of them. I document that sanctions have increasingly targeted firms rather than entire sectors, a substitution unexplained by geopolitical factors. I develop a two-country model of sanctions with heterogeneous firms, international input sourcing and oligopolistic competition. Strategic interaction with unaffected rivals limits the ability of a targeted firm to pass the sanction cost on to consumers, forcing it to absorb much of the shock through lower profits. A sectoral embargo raises every firm's cost uniformly, making a larger share of the sanction reach consumers. In firm-level design, the profit loss relative to the loss in consumer surplus is maximized when targeting the largest firm in a sector. Simulations of the general equilibrium show that firm-level sanctions cost the target economy four times what they cost the sanctioning economy, against eleven under sectoral embargoes. A government maximizing damage inflicted per unit borne would therefore choose the sectoral design. Policymakers nonetheless use firm-level sanctions, revealing a preference over who within the target economy bears the cost.

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1 Introduction

Governments increasingly use economic sanctions as a tool to coerce a target economy (Clayton et al., 2026a; 2026b, 2026c; Mayer et al. 2025, 2026). Sanctions take different forms. Some embargoes ban trade for an entire sector such as with sanctions on arms, oil, or semiconductors respectively in Iran, Russia and China. Other embargoes target specific firms such as Huawei in China, Gazprom in Russia, or Bombardier in Canada. These national champions account for a disproportionate share of national exports and imports. Sanctioning them can therefore inflict substantial damage on the sanctioning and the target economies. Recent examples show an increasing reliance on firm-level sanctions rather than sectoral ones.

This paper studies the incidence of firm-level and sectoral embargoes along two dimensions. The microeconomic incidence is the division of the cost between firms and consumers inside each economy. The macroeconomic incidence is the asymmetry of aggregate welfare losses between the target and the sanctioning economy. I combine reduced-form evidence with a quantitative trade model to estimate how the scope of sanction impacts both dimension of incidence and which incidence matters for policymakers. The scope of a sanction determines the repartition of costs within the economies: consumers pay for a sectoral sanction, whereas a sanctioned firm pays for its firm-level targeting, because competition with unsanctioned rivals forces it to internalize the sanction shock. The macroeconomic incidence runs the other way. A sectoral sanction inflicts on average eleven times more damage on the target economy than it imposes on the economy enforcing it, against four times for a firm-level one. Policymakers nonetheless enforce ever more firm-level sanctions, revealing a preference over who inside the target economy bears the cost rather than over how much damage is inflicted.

Empirically, I use the Global Sanctions Data Base (GSDB) to show that firm-level sanctions accounted for 43% of new embargo cases in 2015 and account for 64% today. I then show that the shift is not a by-product of the geopolitical conjuncture. I run a statistical decomposition fixing, or not, the share of firm-level and sectoral cases conditional on the political objective to predict the counterfactual number of sanction cases by type. Counterfactual scenarios systematically under-predict firm-level cases and over-predict sectoral ones. No discrepancy arises when studying cases encompassing firm-level and sectoral sanctions. The shift is then a substitution from sectoral to firm-level instruments rather than a move toward using them together. Repeating the exercise conditional on political success yields the same conclusion. I then assemble from the archives of the European Union (EU) a dataset of every firm-level sanction since 2001.[1] Matching it to ORBIS, I show that since 2015 the EU has increasingly sanctioned firms in the upper tail of

[1] Montes (2026) sets out its construction and establishes its external validity against the GSDB along both political and economic dimensions.

turnover, employment and assets. At a more aggregated scale, the two instruments target different sectors. Firm-level sanctions hit sectors in the sixth and seventh deciles of import penetration into the EU, while sectoral sanctions, in the sectoral data of Egorov et al. (2025), fall predominantly on sectors with the highest export penetration into the target economy. However an economic sanction acts on both the import and export margins simultaneously. To assess the geoeconomic consequences of firm-level and sectoral sanctions, I therefore embed these two components in a model of international trade.

The model features two economies, each with a finite number of heterogeneous firms competing oligopolistically in every sector, as in Gaubert and Itskhoki (2021). Firms produce using labour and a bundle of inputs from all sectors, including their own, sourced either at home or abroad according to relative prices, as in Amiti et al. (2014). Because firms are large, they internalize their effect on the price of the composite good of their own sector, so their markups respond to both firm-level and sectoral shocks. Importantly, the input sourcing of firms directly enters their pricing decisions. A fully prohibitive sanction encompasses two simultaneous policy shocks. The import ban deprives the sanctioned firm(s) of the demand for final goods and for intermediates in the sanctioning economy. The export control severs the access of targeted firm(s) to inputs abroad and forces them to source within the target economy. The two components therefore hit pricing decisions in both economies and propagate to other sectors in general equilibrium.

The scope of a sanction determines its microeconomic incidence because it dictates whether the shock introduces heterogeneity in producing costs. In the sanctioning economy, a firm-level measure is nested within a sectoral one: imports from the target(s) are prohibited, the remaining firms divide the newly available market share, their profits rise and the price of the composite good increases, thus consumer surplus falls by an amount that scales with the share of the target firm prior to the sanction. Yet, the two measures are not nested in the target economy. An export control against a single firm raises its cost, whereas the cost of its competitors remains unchanged. Passing the sanction cost into its price would decrease its market share, so the sanctioned firm absorbs the greater part of the shock in its markup. Its competitors raise their own prices only to the extent that the designated firm raises its, so the composite price barely moves, and consumers are largely spared. An export control against an entire sector raises every cost homogeneously, no firm experiences a decrease in productivity relative to its competitors, and the sanction cost is mostly passed through into final prices. Pass-through remains incomplete even then, since sanctioned firms still compete with exporters from the sanctioning economy, but also compete to serve intermediate demand in their own economy. I show analytically that pass-through rises with the number of firms experiencing the export control.

To obtain comparative statics on the microeconomic incidence, I compute the precision of a sanction defined as the profit loss of sanctioned firm(s) relative to the surplus loss borne by

consumers in the target economy. I demonstrate that the precision of a sanction is governed by a sectoral quantity comparing the size of intermediate and final demands served, but also the markup and the pass-through of the designated firms. Three comparative statics follow. First, the precision of sanctions increases with the share of intermediate demand in sectoral expenditure. The more upstream the sectors targeted, the more the firms absorb the shock alongside the supply chain towards final consumers.
Second, the pass-through channel dominates the markup one: the sanction is more precise when the target company has the largest market share in its sector. The largest firms charge the highest markups, adjust them most after an export control, and therefore internalize most of the burden of the sanction. Firm-level sanctions weaponize the strategic interaction of firms to increase the precision of sanctions.
Third, targeting a firm that sources more of its inputs abroad raises the loss in markup it bears, but also raises the loss borne by consumers in the target economy by more, so the measure is less precise. It illustrates a tradeoff between hitting the firm's profit harder but seeing more of it passed-through to the final consumer.

I calibrate the model and compare the equilibrium following a sanction to a free-trade baseline. When studying firm-level sanctions, I simulate that the most productive firm in the sector is targeted, as the data indicate. Three results follow. First, a sanction on a single firm costs on average the sanctioning economy 0.14% of welfare and the target economy an average of 0.62%, a ratio of four to one. On average, a sanction on an entire sector costs 0.3% against 3.5%, a ratio of eleven to one. The macroeconomic incidence a sectoral measure generates is 2.7 times that of a firm-level one. If policymakers were interested in the damage inflicted on the target economy relative to the sanctioning economy, they should prioritize sectoral sanctions.
Second, the averaged figures hide substantial heterogeneity across sectors, and even gains in the sanctioning economy in some of them. This occurs in sectors where the targeted firm(s) held a small share of the sanctioning economy's market prior to the sanction. The gain is a rent: domestic firms collect it at home through the import ban, and in the target economy's market through the strategic price increases permitted by the increase of cost for the targeted firms. A model without international input sourcing misses this reallocation of rents altogether and overstates the welfare effect of a firm-level import ban.
Third, contrasting the simulations with the sanctions of the EU against Russia since 2022 shows that its choices are uneven. The EU sectoral embargoes hit sectors that cost its own economy the most but put the largest macroeconomic incidence on the target economy. Its firm-level measures fall on sectors where its own economy gains and the macroeconomic incidence is not the highest. Conditional on enforcing a sectoral measure, the EU selects the most coercive sectors; this is not the case for firm-level sanctions.

The two dimensions of incidence thus deliver opposite prescriptions for scopes. If the objective is to concentrate most of the damage on a firm rather than on the population of the target economy: the sanction should be firm-level. The ideal target to maximize the precision of the damage is a firm with a high market share, and a low share of imported inputs, operating in an upstream sector. If the purpose is to inflict the most damage per unit borne at home: the sanction should be sectoral. The ideal targets are then sectors with the highest import penetration in the sanctioning economy. Governments have predominantly chosen the firm-level sanctions, revealing a preference for who inside the target economy bears that damage. This finding grounds a geoeconomic recommendation that has so far rested on humanitarian grounds: since 2017, jurisprudence in the United Nations has urged policymakers to target sanctions narrowly in order to spare civilian populations. A firm-level sanction spares the population not because it is gentler in aggregate, but because competition from unaffected rivals places most of the sanction burden on the targeted firm.

Related literature. This paper contributes to four key strands of the literature.

First, it contributes to the mostly empirical literature on the economic cost of sanctions. The empirical branch has established that sanctions are costly for both economies. The magnitude and distribution of the losses in the target economy following a sectoral sanction have been extensively documented (Felbermayr et al., 2020; Larch et al., 2021; Flach et al., 2024; Chupilkin et al., 2025; Aytun et al., 2025; Egorov et al., 2025), but much less so is the cost for firm-level sanctions (Nigmatulina, 2025). The cost does not fall on the target economy alone, and the sanctioning economy also bears some costs when imposing sectoral sanctions (Crozet and Hinz, 2020; Crozet et al., 2021), or firm-level sanctions, including in collateral sectors (Besedeš et al., 2021; Besedeš et al., 2024).

The theoretical branch mostly studied how economic sanctions coerce the target economy (Eaton and Engers, 1992; Kaempfer and Lowenberg, 1988) or when a firm-level sanction may be undermined by the government of the target economy (Ahn and Ludema, 2020). This literature has mostly looked at one side of the conflict at a time, either the target or the sanctioning economy. I draw on the literature on geoeconomics and evaluate the macroeconomic incidence of both firm-level and sectoral sanctions on the two economies jointly, showing that a sanction can leave the sanctioning economy better off. I also disaggregate the evaluation of the costs and characterize the repartition of losses within the economies.

Second, this paper relates to the emerging literature on geoeconomics, which studies trade and finance as instruments of coercion (Clayton et al., 2026a, 2026b, 2026c; Mattoo et al., 2026; Becko and O'Connor, 2026; Broner et al., 2025; Mayer et al., 2025; Mayer et al., 2026). That literature focuses on macroeconomic incidence by looking at which economy relies on which, and how much leverage that asymmetry confers. These papers highlight the strategic interactions

of governments to coerce one another. My paper examines the importance of microeconomic incidence to study how governments can strategically weaponize the strategic interaction of firms. Leveraging competition or not modifies the distribution of the costs inside the target economy.

Third, this paper relates to the literature on optimal trade policy. A subset of the literature has examined optimal taxes and subsidies maximizing domestic welfare (Felbermayr et al., 2013; Costinot et al., 2020; Becko, 2024). More recent contributions have examined the optimal tariffs corresponding to a given willingness to pay in a conflict (De Souza et al., 2024). These papers study interior instruments: a tariff that restricts trade while still collecting revenue on what continues to flow. In these papers the revenue incidence generates net gains, outweighing the trade distortion induced. A fully prohibitive sanction sits at the corner where that revenue is zero, so the gains those papers identify are unavailable by construction, and only rent-shifting remains as a potential geoeconomic gain. My paper thus studies in isolation a mechanism that is shut down in the literature because most papers assume constant demand elasticities, which hold markups fixed, thus forcing rent-shifting to be zero.

Fourth, the literature on large firms in international trade established that idiosyncratic shocks to individual firms move aggregates of an economy (Gabaix, 2011; Di Giovanni et al., 2014; Grassi, 2017; Carvalho and Grassi, 2019; Gaubert and Itskhoki, 2021; Baqaee and Farhi, 2024). It also demonstrated that bigger firms absorb more international policy shocks (Di Giovanni et al., 2024; Amiti et al., 2014; Amiti et al., 2019). The closest paper in the literature is Gaubert et al. (2021): they show that tariffs should target large firms in a sector to minimize the pass through into final consumers. I show this prediction survives when incorporating input-output linkages across firms, and abstracting from the tariff revenue. More generally, these theoretical frameworks typically embed a combination of two out of three features needed to study incidence in sanctions: endogenous markup adjustments, international supply chains, and general equilibrium. These features are respectively necessary to study the microeconomic incidence, the export control component, and the macroeconomic incidence of sanctions. Table 5 in Appendix A.1 proposes a summary of the features of the closest models in the literature. I thus contribute by combining these frameworks together, explicitly incorporating international sourcing decisions in pricing decisions of firms to show that large firms are natural targets of a geoeconomic conflict. The reason is now the incidence of the measure inside the target economy, rather than the pass-through calculus of a revenue-collecting tariff.

The remainder of this article is organized as follows. Section 2 presents the datasets used and context. Section 3 establishes motivating facts contrasting sectoral sanctions to firm-level ones. Section 4 presents the theoretical framework and characterizes microeconomic incidence. Section 5 calibrates the model and studies the macroeconomic incidence. Section **??** explores the robustness of the results. Section 6 concludes.

2 Data and context

I assembled data from a wide range of sources to measure economic sanctions and disaggregate the targets within sanction episodes. This section describes the key datasets I rely on, with Appendix A providing additional details.

2.1 Dataset on worldwide sanctions

To distinguish between firm-level and sectoral sanctions, I study sanction cases. Sanction cases are legal texts, published by the countries to enforce an economic sanction. The Global Sanction DataBase (GSDB) by Syropoulos et al., 2024[2] classifies the cases enforced between 1949 and 2025 from one country to another. Across the period, 1,794 cases have been imposed, 828 of which are still active today. The number of active sanctions has been multiplied by four in the last 15 years (see appendix figure 5 for an illustration). The sanctions span the entire world. Appendix chart 6 shows that, in 2025, around 140 countries are targeted by sanctions, and 175 countries have ever been sanctioned. In contrast, only a few countries impose sanctions: mostly the USA and Europe (see appendix figures 7 and 8 for respective maps of target and sanctioning economies).

Sanctions can have different political objectives, grouped by the authors in the following categories: prevent war, end war, territorial conflict, policy change, democracy, human rights, destabilize regime, terrorism, other. The political success is inferred comparing *ex ante* stated policy objective to realized political states. They group the political success in 5 categories: total, partial, negotiation, ongoing, failed. A given sanction can have several political objectives and thus has several success evaluations associated. I consider different political objectives and success as being independent in the following exercises.

Sanctions can be of different type: each banning economic resources to different targets in an economy. The GSDB decomposes every sanction case in 6 types: trade, arms, military, financial, travel and a category other types of sanctions. A sanction case can be of multiple types at the same time.

Trade sanctions restrain economic interactions with a target country by limiting international trade. Trade sanctions can be either partial, targeting only imports, or only exports, or complete. The main results of this paper focuses on complete sanctions since partial firm-level sanctions do not exist. However, partial sanctions can also be studied within the model. In practice, the oil embargo against Russia, or the export control against the semi-conductor industry respectively enter the complete trade sanction, and partial trade sanction categories. Arms sanc-

[2]See also Felbermayr et al., 2020 and Kirikakha et al., 2021

tions restrict arms sales. Sanctions on military assistance ban monetary or personal assistance in the target country. Practically, these three types of sanctions designate some product code(s) and prohibit any economic interaction of domestic producers and consumers with the goods designated[3]. They thus apply uniformly to all foreign firms operating in the product code, called Harmonized System (HS) code, targeted. I thus classify as sectoral sanction a sanction case whose type is either trade, arms, or military. There are currently 88 programs actually enforcing only sectoral sanctions, which represents 11% of the total sanction cases active today.

Financial sanctions freeze the asset, restrict direct investments and/or limit the availability of credit for payments in the exchange of commodities for certain entities and/or individuals. They provide lists of names and prohibit making any economic funds available to the targeted entities. In practice Huawei enters the GSDB through a program of financial sanctions imposed by the USA against China, and Gazprom as well in the financial sanctions enforced by the EU. Travel restrictions restrict the freedom of geographical movement of individuals. The other category is residual and primarily entails diplomatic measures, as well as flight and harbor restrictions.
European sanction cases help guide intuitions on the scope and targets of these targeted sanctions. Practically, in most cases European travel bans target persons for their work within a sanctioned entity. Appendix A.3 provides anecdotal evidence showing that across reasons, it is easy to find sanction motivated by the work of the targeted individual.
I classify as firm-level a sanction case that is either financial, travel or *others*. There are currently 342 programs actually enforcing firm-level sanctions, which represents 44% of the total sanction cases active today. All results shown in this paper still hold when classifying *other* sanctions into the sectoral category. If a sanction case is encompassing both firm-level and sectoral sanctions at the same time, I classify it as a mixture of both. In the GSDB, 670 cases are a mixture, 59% of which are still active today. The results are amplified when excluding mixture cases from the sample.

2.2 Describing the targets of a sanction case

Though the GSDB covers the sanctions enforced by all countries in recent years, it does not detail the precise firms and sectors sanctioned. Recent datasets have been developed, focusing on specific sanctioning economies or target economies, to fill this gap.

Sectoral sanctions. To precisely identify the sectors sanctioned, Egorov et al. (2025) focus on the sanctions imposed against Russia following its invasion of Ukraine in February 2022. Their

[3]The EU imposes all types of sectoral sanctions. Reg 833/2014 encompasses all possible types: complete trade sanctions such as 9 CN codes for semiconductor devices, among which "CN 8541 10: Diodes, other than photosensitive or light-emitting diodes (LED)" (Annex VII, Part B), partial trade sanctions seen as: "Clock with a frequency that exceeds 25 MHz" (Annex VII, item X.A.I.001.a), arm sanctions as in: "CN 9303: Other firearms and similar devices which operate by the firing of an explosive charge" (Annex XXXV of Article 2aa).

dataset encompasses the sanctions enforced by Australia, Canada, the European Union (EU), Japan, Korea, Switzerland, Taiwan, the United Kingdom, and the United States. They provide with the staggered dates at which the sanction started being enforced from February 2022 to July 2024.

To attribute which sector is sanctioned, they either pick the product code listed in the text case, or assign product codes based on the textual description of goods made by the governments. They do not distinguish between partial and complete bans, assuming they are all fully prohibitively sanctioned and impact indifferently all firms in this sector.

Firm-level sanctions. To precisely identify the firms sanctioned, I compile a dataset of every financial sanctions imposed by the EU since the creation of the consolidated lists of the Council of the European Union. I complete and correct the public list of entities thanks to information gathered in the archives of the official journal of the EU.

The database covers 24 sanction programs enforced between 2001 and July 2025. Each of those 24 sanction cases is recorded in the GSDB, with famous examples including Russia, Iran, North Korea, etc. In Montes (2026) I demonstrate its external validity both in political and economic dimensions, relative to the GSDB trends. I disaggregate the targets of those 24 firm-level programs into $3,270$ entities that were, are or have been sanctioned by the EU. 65% of the entities are not sanctioned anymore today, which illustrates that public lists currently available online miss two-thirds of all entities ever targeted. My dataset also precisely tracks the period of the threat of sanction, the starting point of the sanction, and the lift of sanction. For most entities, I gather the precise address(es) of all establishments targeted by the EU.

An entity in my dataset can be a firm, but also a ministry or a research center. In Montes (2026), I match it with ORBIS to precisely identify firms. ORBIS by Bureau Van Dijk is a world-wide dataset providing financial information on firms, gathered through government information or public listings. The match is done on the exact name of the firm and its exact address measured at the city $\times$ country level. It yields 1,207 firms sanctioned, 584 of which are still sanctioned today.

On average 67 firms are included per firm-level program. A back-of-the-envelope computation, assuming that all countries apply similar firm-level sanctions as the EU, estimates around 22,914 firm-level sanctioned companies worldwide in 2025. In practice, the USA has fewer legal requirements to enforce sanctions and this computation is likely to be a lower bound.

Firm-level targets can be multi-product firms. It thus follows that even more sectors may be impacted by firm-level sanctions. On average, 824 HS6 codes are impacted per program. Comparing the two averages yields firms operating in roughly 12 HS6 codes. Once aggregating at the HS2 level, there are, on average, 24 distinctive sectors impacted per sanction program. A back-of-the-envelope computation similar to before, multiplying the 342 firm-level programs by

the average 824 HS codes per program, estimates around 281,808 HS6 sectors between country pairs impacted by firm-level sanctions in 2025.

2.3 Other trade datasets

Sectoral trade flows. To characterize which sectors are targeted by sanctions, I use bilateral trade flows at the 6 digits level. BACI, developed by Gaulier and Zignago (2010) by CEPII gathers such data at a yearly frequency from 2002 to 2025. This dataset is convenient when studying economic sanctions because CEPII is maintaining a trustworthiness score for the declarations of every country. Intuitively, the export from France to Russia should be equal to the import of Russia from France, after correcting for differences in the reporting of trade costs. They correct for reporting discrepancies, and reweigh the two reported trade flows by the confidence index to obtain the figure reported in their dataset. This reweighting can prove useful when studying countries, already sanctioned in the past, trying to manipulate their declared trade flows to hide circumventions. I use the dataset in its HS 2017 nomenclature and mainly use bilateral trade flows between countries from the EU and Russia in 2021. Focusing on the EU follows from the scope of the firm-level sanction dataset. Focusing on Russia as the target economy follows from the scope of the sectoral sanction dataset. The baseline year is chosen as the year before February 2022, which corresponds to the beginning of the full scale invasion of Ukraine by Russia.

Input-output tables. The empirical exercises cannot account for the input-output of firms and is thus constrained to study only one component of economic sanctions, neglecting the spillovers to other sectors. To circumvent this limitation, I build and estimate a model of economic sanctions with supply chains, that I calibrate on the World Input Output Database (WIOD), compiled by Timmer et al. (2015). This database integrates national input-output tables with data on bilateral trade flows to construct a comprehensive matrix detailing all intranational and international flows of goods and services between sectors and final consumers. I use the 2016 release of the WIOD, which encompasses 43 countries (plus an aggregate for the rest of the world) and 56 sectors categorized according to the ISIC Revision 4 classification. The WIOD dataset is instrumental in estimating: i) the sectoral income expenditure shares; ii) the shares of labor and intermediate inputs in production functions; and iii) the shares of imported inputs.

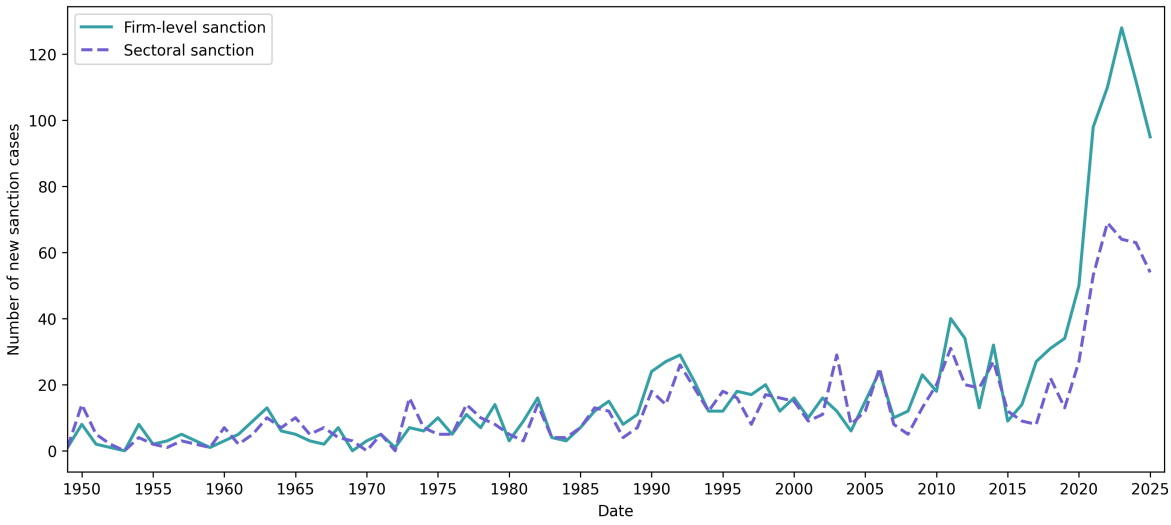
3 Motivating facts

The granular datasets I have collected allow me to describe the use and scope of sectoral vs. firm-level sanctions at different macroeconomic levels. In this section, I present new facts describing how policymakers enforce sanctions and reduced-form evidence on the effect of geopolitical out-

comes on firm-level vs. sectoral sanctions. I report additional descriptive statistics in Appendix B.

Fact 1: new sanction cases mainly target specific firms rather than sectors. Using the GSDB, I represent in figure 1 the enforcement of new firm-level and sectoral sanction cases across time. A sanction case mixing firm-level and sectoral sanctions is counted as one in each category. In total, accounting for the cases with firm-level and sectoral sanction, 43% of new sanction cases in 2015 were firm-level, this share has grown to 64% today. Some of those embargoes may have ended, but in terms of active sanctions the cases with some firm-level component represented 72% in 2015 and now represent 92%. Similarly, The number of firm-level sanctions is increasing faster than the enforcement of sector-wide ones, and this is not counterbalanced by higher lift of firm-level sanctions. The count for active sanctions, separating firm-level only, sectoral only, and a mixture of the two can be found in appendix figure 9. It further describes that the increase is mainly driven by purely firm-level sanctions, but also supported to a lesser extent by a mixture of sectoral and firm-level sanctions. Altogether, these graphs confirm the sharp increase in sanction enforcement discussed by the literature (GSDB), but it contributes by showing that the main driver is firm-level sanctions.

Figure 1: Evolution of the Enforcement of New Sanction Cases per Year



Note: Firm-level and sectoral sanctions are inferred from the typology of sanctions of the GSDB (Syropoulos et al., 2024). The graph highlights that firm-level sanction is the main driver of the novel increasing use of sanctions. A sanction encompassing firm-level and sectoral sanctions is counted as one in both curves.

However a sanction program may sanction different numbers of sectors and firms. It may be that the texts enforcing firm-level sanctions target very few firms when texts for sectoral sanctions have many targets. In the aggregate, sectoral sanctions would then predominate and outweigh the observed shift in figure 1. To address this concern, I compare the staggered enforcement of sectoral and firm-level sanctions against Russia. To compare the two, I assume that an HS6 code is impacted by a sanction when at least one firm in that sector is sanctioned. The appendix figure

10 shows that a sanction case enforcing sectoral and firm-level sanctions impacts a similar number of sectors. The graph also shows that the sectors impacted by firm-level sanctions and sectoral sanctions are mostly not the same. The overlapping sectors may be due to the assumption of Egorov et al. (2025), not distinguishing between partial and complete sectoral sanctions. This is reassuring since both types of sanctions should no co-exist in a same sector. The appendix figure 11 uses the FSDB to its full extent and further shows that the number of firms sanctioned by the EU is globally increasing. All in all firm-level sanctions are increasingly used, targeting more firms, producing in different sectors thus impacting more and more sectors.

Fact 2: conditional on any geopolitical outcomes, policymakers have substituted from sectoral to firm-level sanctions. An additional concern may be that this new prevalence of firm-level sanctions is entirely driven by geopolitical objectives. The relative increase of firm-level sanction with respect to sectoral ones may come from three different sources: a scale effect, an increase in the use of a political objective only using firm-level sanction, or a shift in the pattern of imposing sanctions. Respectively, first, the scale effect may be a mechanical consequence of the increase in total number of cases. Assume an asymmetry in the type of sanctions used ex ante where firm-level sanctions were slightly more used. The total increase of sanctions may accentuate this contrast and create that new prevalence of firm-level sanctions. Second, one could think that, across the 9 different objectives available in the typology of the GSDB, some are using purely firm-level while some others only sectoral level, maybe due to uncaptured enforcement costs. The increase could then just follow a shift in geopolitics, enforcing sanctions on one political objective relying exclusively on firm-level sanctions. Last, one could see a shift in the way policymakers impose sanctions, going from sectoral cases to firm-level cases to tailor the targets and reduce the economic costs. Put otherwise, this third channel considers a substitution from one type of sanction to the other, conditional on geopolitics.

To disentangle the relative importance of those three channels, I statistically decompose the number of sanctions by type, following Shapiro and Walker (2018). To demonstrate that the increase in the firm-level sanctions relative to sectoral ones comes from the substitution, rather than the complementarity of firm-level to sectoral sanctions. To do so, I study three types of sanctions: cases with solely firm-level sanctions, solely sectoral sanctions, or cases with a mixture of both. The first two allow me to study the substitutability across sanction types, while the latter rather documents whether sanction types have become more complementary. For each of those three types, I study the following shift-share decomposition:

$$\underbrace{SC_t(type)}_{\text{Nb case by type}} = SC_t \times \sum_{objective} \underbrace{\theta_t(objective)}_{\text{Share of political obj}} \times \underbrace{\nu_t(type, objective)}_{\text{Share of type in pol. obj.}} \tag{1}$$

where a type can be either firm-level, sectoral or a mixture of both, $SC_t(type)$ is the total number of active Sanction Case of a given type at date t, SC_t the total number of active Sanction Case at date t irrespective of the type, $\theta_t(objective)$ the share of active sanction with a political *objective* at date t, and $\nu_t(type, objective)$ the share of active sanction of a given type, within an objective o, at date t. The appendix graph 13 captures sequentially the relative importance of the scale, the political objectives, and the type within political objective of sanctions, in the relative increase of firm-level to sectoral sanctions.

Empirically, for every type of sanction, I first hold fixed the shares of objectives and the share of type (i.e. $\sum \theta(objective)\nu(type, objective)$) to their value of 2010[4], to represent the green curve in Figure 13. In that counterfactual, the structure of sanction does not move since before the observed asymmetry, thus only the scale changes. Second, I keep fixed only the share of type inside a political objective (i.e. $\nu(type, objective)$) to its 2010 level. The blue curve then captures the dynamic change of both the scale and the patterns in political objective. Finally, I let all elements of the decomposition vary to obtain the observed count of active sanctions by type, represented in red in Figure 13. The red curve thus encompasses both the variation of the scale, the pattern of political objective, and the evolution of the sanction type within objective.

To observe the source of the asymmetry, one needs to study the decomposition for the three types of sanctions. If the asymmetry was explained by the increase in the mixture of the two, it should reflect an increasing complementarity of sanction types. If the asymmetry is explained simultaneously within objectives by a decrease of sectoral level and an increase of firm-level ones, then the two types of sanctions have become more substitutable.

The figure 13 shows two main messages. First, the graph at the bottom of Figure 13 evaluates equation (1) when considering the cases that encompass both firm-level and sectoral sanctions. It is purely increasing due to a scale effect. Its composition across and inside political objective has not changed since 2010. Thus, the increasing asymmetry does not come from an increase in complementarity between sanction types. Second, the firm-level decomposition at the top of Figure 13 shows that both curves holding fixed the type of sanction within objectives cannot replicate the observed trends in the red curve. The current increase in firm-level sanctions is thus mainly driven by an increase in the use of firm-level sanction conditional on the political objective. The sectoral sanctions are, on the contrary, increasing to a lower extent in practice than in counterfactual scenarios because of the change of type used conditional on political objective.

This rules out the possible story of political objectives specialized in firm-level sanctions being increasingly used, since it would be translated into a gap explained by the blue curve, which should then coincide with the red curve. Combining those two analyses, it follows that the asymmetry observed in figure 1 comes from a reallocation of shares inside political objective from sectoral

[4]Results are robust to setting the year-value to 2000.

sanction to firm-level ones. Policymakers are thus substituting away from sectoral sanctions to firm-level ones.

Finally, changing geopolitical outcome, it may be that this increase of firm-level sanctions relative to sectoral ones reflects a higher success rate of firm-level sanctions. Doing the same exercise but focusing on success rate rather than political objectives provides similar conclusions. For both exercises, the results are robust to excluding Russia or China from the target. The results also hold when excluding the USA, or Europe from the sanctioning state. Appendix Q.1 presents the robustness tests.

It then follows that firm-level sanctions have become prevalent conditional on political objectives, and success rate. Given that geopolitics does not rationalize the shift, it follows that geoeconomic incentives must be at play. I characterize the targets of European sectoral and firm-level sanctions to inform important dimensions of the quantitative exercise.

Fact 3: large firms represent a high share of firm-level targets. Since geoeconomics is the next candidate to explain this shift, I describe the economic characteristics of the targets of sanctions. Using the merge between FSDB and ORBIS, I can observe the size classification created by ORBIS for every firm-level sanctioned company. They classify large-size firms as those satisfying at least one of these conditions: i) having an operating revenue greater than €10 million, ii) having a total stock of assets greater or equal to €20 million, iii) having 150 or more employees. Very large firms are defined similarly, when multiplying by 10 each of the three thresholds. Appendix figure 14 shows that since 2015 the share of large firms has increased from 23% to 40%. It follows that large firms have become an important component of firm-level targets of the European sanctions.

Fact 4: sectoral sanctions target sectors with high export penetration and firm-level sanctions rather target sectors with average import penetration. To get an intuition at a different scale on the targets of sanctions, I study how important the sanctioned sectors are for the bilateral trade flows between the EU and Russia. I can only measure one side of bilateral flows for both sectoral and firm-level sanctions given that the input-output structure of firms is not observable. For sectoral sanctions, I observe the HS code of the sanctioned sector, I thus look at the distribution of exports from the EU to Russia and bin sectors by their export values. Inside of every bin, I compute the share of sanctioned sectors. The highest bars in appendix figure 15 thus represent how much export value the EU is willing to give up when doing sectoral sanctions. The graph shows an increasing share of sanctioned sectors as the export penetration increases. The EU thus primarily targets sectors with bigger trade flows. The firm-level sanctions allow me to analyze the other side of the bilateral flow: import penetration of Russian exports into the EU market. Making the assumption that a firm-level sanction affects the full sector provides an upper bound of the reach of sanction, but allows me to detect the sectors targeted by the EU. Again, I cannot

study exports from the EU since the supply chain of Russian firms is not observed. I find in appendix figure 16 that the EU sanctions a lower share of sectors it is importing from. Additionally, the EU predominantly targeted sectors with average import penetration, in between the fifth and eighth decile of import penetration.

Remaining questions. The evidence presented in this section highlights the new prevalence of firm-level sanctions conditional on geopolitical outcomes. However, it is unclear how is that new scope driven by geoeconomic rationale. Additionally, reduced-form evidence on incidence is not possible for some reasons: i) the observable sanctions result from an endogenous voting of European countries, ii) sanctions cannot be fully captured in reduced form since it would require the full input-output structure of firms, iii) sanctions need to be evaluated in general equilibrium as they may spillover from one sector to another and affect terms of trade. All those reasons drive me to build a model of international economic sanctions, to answer key policy questions: is there any rationale for targeting more narrowly? Which firms should be targeted by firm-level sanctions? Which sectors, in firm-level and sectoral sanctions?

4 A model of economic sanctions

The framework is a granular trade model with oligopolistic competition based on Gaubert et al. (2021) to which I incorporate inputs import decisions by firms à la Amiti et al. (2014). The model features both sectoral comparative advantage and granular comparative advantage within sectors, arising from idiosyncratic productivity draws of individual firms. In the model, governments only have one policy tool: sanctions. The domestic economy enforces sanctions against the foreign one, without any retaliation. This assumption omits 5% of sanction cases in the GSDB. This is a moderate assumption, when abstracting from tariffs.

4.1 Theoretical Framework

The world consists of two economies: Domestic and Foreign. The Foreign economy represents the rest of the world and is indexed by a star. It is modeled symmetrically to Domestic: the preferences, technologies and elasticities are identical. In the absence of sanction, countries freely trade composite goods. Trade in final goods is costly. There is a continuum of sectors, denoted z or k, included in $[0, 1]$. The sector z^* is the analogous of the domestic sector z in the foreign economy. From now on, firms producing final goods are denoted by i or j.
The model considers a sanction against a foreign firm j^*, producing in the foreign economy, operating in the foreign sector z^* and exporting to the domestic economy in the sector z.

Representative household. There are L (resp. L^*) households in Domestic (resp. Foreign),

inelastically supplying labor. They have Cobb-Douglas preferences in their consumption in all sectors C_z, with expenditure shares α_z. Utility is ordinal: I assume without loss of generality that $\int_0^1 \alpha_z dz = 1$.

$$U = \exp(\int_0^1 \alpha_z \log(C_z) dz)$$

The maximization of the utility yields that the household spends a constant share of her income I in a sector z: $P_z C_z = \alpha_z I$. It follows that the aggregate price index is:

$$P = \exp\left[\int_0^1 \alpha_z \log\left(\frac{P_z}{\alpha_z}\right) dz\right] \tag{2}$$

Set of firms in a market. There is in each sector z a finite exogenous number of firms each producing only one final good. Multiproduct firms must be thought of as several distinct firms in this framework. I call $\mathcal{D}_z$ the set of domestic final good producers, and $\mathcal{F}_z$ its foreign analog. Economic sanctions constrain these sets in both locations. I then denote $\mathcal{A}_z(\beta)$ the set of admissible sellers from which agent β in an economy, be it a consumer or a firm can buy. The set of all admissible firms from domestic in a market is $\mathcal{A}_z \cap \mathcal{D}_z$. The set of all admissible firms from foreign in a market is $\mathcal{A}_z \cap \mathcal{F}_z$. In the domestic market, the set of all possible sellers to the agent β, denoted $\Omega_z(\beta)$ is $\Omega_z(\beta) \equiv (\mathcal{A}_z(\beta) \cap \mathcal{D}_z) \cup (\mathcal{A}_z(\beta) \cap \mathcal{F}_z)$. Intuitively it corresponds to all domestic firms and foreign exporting firms in the domestic market z. The first term in parenthesis captures export control, and the second one captures import ban.

An import ban against a firm j^* in z^* excludes j^* from $\mathcal{A}_z(\beta)$ for all agents β in the domestic economy (i.e. firms and consumers). An export control against j^* in z^* imposes that $\Omega_z(j^*) = \varnothing$. Since the subscript is indexed by z and not z^* it means that the composite good is produced in the domestic economy, and j^* can thus no longer use it to produce.

Production of composite goods. Sectors are comprised of heterogeneous firms. Each horizontally differentiated variety is aggregated in a CES function $Q_z = \left[\sum_{i \in \Omega_z} q_{z,i}^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}$ where σ represents the constant elasticity of substitution across varieties, assumed identical across sectors and strictly greater than 1, and $q_{z,i}$ is the quantity produced by a firm i in z. The program of the composite good producer yields:

$$q_{z,i} = \left[\frac{p_{z,i}}{P_z}\right]^{-\sigma} Q_z \tag{3}$$

where I denote the price of the composite good of z:

$$P_z \equiv \left(\sum_{i \in \Omega_z} p_{z,i}^{1-\sigma}\right)^{\frac{1}{1-\sigma}} \tag{4}$$

Equation (3) characterizes the demand faced by firm i producing its final good in sector z. Equation (4) defines the price of the composite good in a sector z produced domestically. The composite good is respectively consumed by domestic consumers or used as an input by firms producing final goods in both domestic and foreign economies: $Q_z = C_z + M_z$.

Production of final goods. Each firm faces the downward sloping demand given by (3) in its distinct product variety. Their constant return to scale production technology mixes labor ($\ell_{z,i}$) with a bundle of composite inputs k sourced either domestically in sector k or from the same sector, in Foreign economy k^*: $({M_{z,i}}^{k,k^*})$. Labor is supplied inelastically by domestic workers. I then assume that the production function of a firm i in sector z is $y_{z,i} = \ell_{z,i}^{\gamma_z} \exp\left[\int_0^1 \gamma_z^k \log({M_{z,i}}^{k,k^*})dk\right]$ with γ_z the constant share of total variable cost spent on labor in sector z. γ_z^k represents the share spent on composite good from sector k by firms in sector z. Constant returns to scale ensure that $\int_0^1 \gamma_z^k = 1 - \gamma_z$. Technological parameters are identical across firms in a same sector. However, the input bundle is firm-specific: $\forall k \in [0;1] : {M_{z,i}}^{k,k^*} = [\left(m_{z,i}^k\right)^{\frac{\chi}{1+\chi}} + \left(\rho_z^k\right)^{\frac{1}{1+\chi}} \left(m_{z,i}^{k^*}\right)^{\frac{\chi}{1+\chi}}]^{\frac{1+\chi}{\chi}}$ where $m_{z,i}^k$ is the quantity of domestic composite good k used by firm i in z, and $m_{z,i}^{k^*}$ is the analog quantity but sourced from abroad. The substitutability across locations for a given composite good k depends on the elasticity of substitution across locations $1 + \chi$ assumed constant across sectors, the relative input prices (P_k/P_{k^*}) and the exogenous share of sourcing abroad ρ_z^k, which is constant across firms. In the free-trade benchmark, there is no heterogeneity in sourcing of inputs among firms in a same sector because composite goods are freely traded for every firm.

Entering domestic and foreign markets. At the beginning of the period, the firm draws in a stochastic Pareto distribution a productivity parameter $\varphi_{z,i}$ that reduces its cost in a multiplicative way. Productivity is then fixed for the rest of the period. To enter its domestic market, a firm i in z then has to pay a fixed cost F in labor term. If its productivity allows, the firm enters and serves domestic market by producing in its domestic location. Similar to Atkeson and Burstein (2008), this model does not feature productivity cutoff at the extensive margin for exports. The same applies to imports, contrary to what is done in Amiti et al. (2014). When exporting, firms need to overcome an iceberg trade cost, denoted κ: for each unit sold in the Foreign market, a firm needs to send $\kappa > 1$ units. All firms can export, as soon as they are productive enough to overcome the iceberg trade cost and obtain market share in the foreign market. Their sole productivity is driving the markets they serve.

The cost-minimization program yields (details in Appendix D.1) the unit cost of a firm i in sector z:

$$c_{z,i} = \underbrace{\kappa}_{\substack{\text{Additional cost when} \\ \text{exporting}}} \times \underbrace{\frac{1}{\exp\left(\int_0^1 \gamma_z^k \log(b_{z,i}^k) dk\right)}}_{\substack{\text{Enhancement of productivity} \\ \text{when sourcing abroad} \equiv B_{z,i}}} \times \underbrace{\frac{1}{\varphi_{z,i}} \left[\frac{w}{\gamma_z}\right]^{\gamma_z} \exp\left(\int_0^1 \gamma_z^k \log(\frac{P_k}{\gamma_z^k}) dk\right)}_{\substack{\text{Unit cost of non-importing} \\ \text{and non-exporting firms} \equiv C_{z,i}^{\mathsf{CP}}}} \tag{5}$$

where I denote w the wage from supplying one unit of labor, and $b_{z,i}^k = \left[1 + \rho_z^k \left(\frac{P_k^*(i)}{P_k}\right)^{-\chi}\right]^{\frac{1}{\chi}}$.

The latter represents the enhancement of productivity of firm i in z created by sourcing abroad the composite good k. This endogenous variable is greater than 1 and decreases in a multiplicative way the unit variable cost. Absent any sanction, this quantity is constant across firms within sector z, for every imported input k. $B_{z,i}$ is thus constant for all firms i in $\mathcal{D}_z$. The sourcing decision directly affects the unit cost through that term.

The third term in equation (5) is the unit cost supported by firms producing with domestic inputs and serving the domestic market. It is similar to the cost derived in Caliendo and Parro (2015). The model therefore directly inherits from the class of models using the roundabout assumption: the unit cost of i in z depends on the composite price P_z, which depends itself on $p_{z,i}$ by equation (4). This assumption is relevant when modeling the energy sector, which suits the case of sanctions often imposed against energy sectors (Russian oil and gas).
This assumption allows to integrate spillover effects of sanctions in the model. In the general equilibrium, once prices of every sector adjust, a sanction in a single sector can propagate to other sectors in the economy. The model thus accounts for collateral damages described in the literature (Hall and Seyfi, 2021; Besedeš et al., 2024).

Impact of a sanction on the unit cost. Economic sanctions act through two channels: export control and import ban. The former impacts the cost of producing of the sanctioned firm in the target economy, meanwhile the latter only impacts the exports of the target.

On the one hand, the import ban forbids any import from firm j^* into the domestic market. It follows that $j^* \notin \Omega_z(\beta)$ for all β in the domestic economy. j^* is then excluded from the composite good aggregation and then does not enter in the computation of P_z in (4). In practice, it reduces the productivity of Foreign firm j^* from z^* to 0 in the domestic market z. The import ban alone only impacts the production of the sanctioned firm through the relative composite prices in the general equilibrium.

On the other hand, the sanction forbids all firms in the domestic economy to export their composite good to firm j^* in z^*. It follows that all domestic sectors $k \in [0; 1]$, are such that, $\Omega_k(j^*) = \varnothing$: j^* must source domestically and thus $\forall k : b_{z^*,j^*}^k = 1$ by definition. The productivity enhancement term in the unit cost of firm j^* in z^* decreases to 1 which in turn increases the cost

of firm j^* when serving the market z^*. Absent any import ban, an export control then increases the target firm's cost to produce in both domestic and foreign markets. Unsanctioned rivals keep having their enhancement of productivity decreasing their unit cost. In other terms, the export control constrains the sourcing strategy of firm j^* and thus introduces heterogeneity in importing with respect to other unsanctioned firms in the market z^*.
In a firm-level sanction, the additional heterogeneity only affects the sanctioned firm, meanwhile a sectoral sanction homogeneously increases the costs of firms. A sectoral sanction does not introduce any heterogeneity compared to a free trade scenario.

The variables $(b^k_{z,i})_{i,(k,z)}$ are a crucial component of sanctions in the model as they break the standard property of separability across markets at the equilibrium. Relaxing this assumption allows that a sanction can not only have an impact on the markets in the sending economy, but also alters competition in the markets of the target economy. When studying the macroeconomic incidence, it is important to relax this assumption to have precise estimates of the welfare change in the target economy.

Strategic interaction of firms. Inside a sector z, firms play oligopolistic competition. Since there is a finite number of firms in the sector, they can internalize their impacts on the sectoral aggregate. Their pricing strategy thus internalizes their impact on composite good price P_z and composite quantities Q_z. The Cobb-Douglas structure of preferences and technologies yields $Q_z = \frac{\alpha_z I}{P_z} + \frac{\Psi_z}{P_z}$ with $\alpha_z I$ the constant share of income spent by the consumer in sector z and Ψ_z the aggregate variable cost spent in the domestic sector z by every firm from Domestic and Foreign. Even if the firm is very productive within its sector, it is infinitesimal in the overall economy. Therefore it takes I and Ψ_z as given in its program. A complete formulation of Ψ_z in Appendix D.2 further details its exogeneity to the program of the firm.

Using equation (3), the expenditure on variety i in sector z can be rewritten

$$p_{z,i}q_{z,i} = \left[\frac{p_{z,i}^{1-\sigma}}{P_z^{-\sigma}}\right](C_z + M_z) = \left[\frac{p_{z,i}^{1-\sigma}}{P_z^{-\sigma}}\right]\frac{1}{P_z}(\alpha_z I + \Psi_z) = s_{z,i}(\alpha_z I + \Psi_z)$$

where the within-sector z market share of the variety i is defined as follows:

$$s_{z,i} = \left(\frac{p_{z,i}}{P_z}\right)^{1-\sigma} \tag{6}$$

Equation (6) expresses that the higher the price of a firm relative to the composite price of its sector, the lower its market share.
When playing *à la Bertrand*, every firm maximizes its profit taking prices of every other firm in

the economy as given. The program is similar but with quantities for Cournot competition. The Nash equilibrium of the Bertrand game is given by the following pricing rule:

$$\frac{p_{z,i}}{c_{z,i}} = \frac{\epsilon_{z,i}}{\epsilon_{z,i}-1} - \frac{\gamma_z^z s_{z,i} (b_{z,i}^z)^{-\chi}}{\epsilon_{z,i}-1} \equiv \mu_{z,i} \tag{7}$$

where $\epsilon_{z,i} = [\sigma + s_{z,i}(1-\sigma)]$ represents the perceived elasticity of final and intermediate demand. The more productive a firm is, the lower its price relative to other firms in the sector, the higher its market share and so the less elastic the part of the demand curve it faces. Equation (7) defines the endogenous markup of firm i when serving market z, denoted $\mu_{z,i}$.

The second term, adding to the standard Lerner formula, originates from the roundabout assumption. It represents the fact that the firm i in z internalizes that increasing its price $p_{z,i}$ raises the price of the composite good of its sector P_z, which in turn increases its production cost $c_{z,i}$. However, the $b_{z,i}^z$ term adds that the more the firm is importing its input from the economy abroad, the less this force applies. In the free trade baseline, this decrease in markup applies uniformly to all firms in the market z and is weighted by the firm reliance in the composite good from its own domestic sector γ_z^z.

Equation (7) clearly shows that the markup of a firm is decreasing in the share of domestic input in the intermediate good bundle $(b_{z,i}^z)^{-\chi}$. Consequently, an export control cuts the supply chain of a firm, forces it to source domestically and experience to a greater extent the internalization of the cost of a firm. The export control thus impacts to a first order, the markup of a sanctioned firm in its own domestic market. Input sourcing is however not the only endogenous variable impacting the markup since the market share also impacts it. An exercise of perturbation around the equilibrium will indicate which impact is stronger, later in this manuscript.

One can reformulate equation (7) and include the Cournot case as follows $\frac{p_{z,i}}{c_{z,i}} = \frac{\xi_{z,i}}{\xi_{z,i}-1}$ where the residual elasticity to intermediate and final demand is:

$$\xi_{z,i} = \begin{cases} \frac{\sigma + s_{z,i}(1-\sigma) - \gamma_z^z s_{z,i} \left(b_{z,i}^z\right)^{-\chi}}{1 - \gamma_z^z s_{z,i} \left(b_{z,i}^z\right)^{-\chi}} & \text{under Bertrand} \\ \frac{\sigma \left[1 - \gamma_z^z s_{z,i} \left(b_{z,i}^z\right)^{-\chi}\right]}{1 + s_{z,i}(\sigma-1) - \sigma\gamma_z^z s_{z,i} \left(b_{z,i}^z\right)^{-\chi}} & \text{under Cournot} \end{cases} \tag{8}$$

From this reformulation, it is straightforward to derive that $\mu_{z,i}$ increases with market shares $s_{z,i}$. The bigger firms in a sector are charging the higher markups.

Equation (8) illustrates clearly the forces endogenously shaping the markup. On the one hand a higher market share implies a higher markup, which allows the firm to set a higher wedge between its cost and price than $\sigma/(\sigma-1)$. The market shares departs from the constant markup set in monopolistic competition.

The internalization of own cost acts in the opposite direction: taking the extreme case of complete internalization (i.e. $\gamma_z^z(b_{z,i}^z)^{-\chi} = 1$), one sees in (8) that the elasticity of the demand converges to σ for both Bertrand and Cournot competition. Such a residual elasticity leads to the markup under monopolistic competition.

In equilibrium, firms with higher market shares command higher profits, and firms with higher enhancement of productivity (i.e. $b_{z,i}^z$) as well.

Partial equilibrium. The Nash equilibrium of a sector z is reached at the crossing of the best-response functions of all firms i in that sector. Formally, taking $(w, (P_k)_{k\neq z}), (P_k^*)_k)$ as given, the equilibrium of sector z is given by the vector $\left(p_{z,i}, b_{z,i}^z, c_{z,i}, s_{z,i}\right)_{i\in\Omega_z}$ such that every firm in z optimizes its input bundle, and maximizes its profit, taking the prices of its competitors as given. Iterating over the sectors z of an economy, the equation (4) directly provides the composite price for every sector at their equilibrium.
Given the sources of heterogeneity in the model, productivity draws drive the *laissez-faire* allocation between firms at the sectoral equilibrium. Productivity draws and sourcing decisions drive the allocation post sanction.

General equilibrium. The expenditure for every sector z is denoted $X_z = P_z Q_z$. The general equilibrium vector contains wages and expenditures for every sector in the world (w, w^*, (X_z, X_{z^*})$_{z\in[0;1]}$) such that labor and good markets clear in both countries and trade is balanced.

The total labor compensation is equal to the sum of all the expenses on labor done by domestic firms.

$$wL = \int_0^1 \left[\sum_{i\in\Omega_z} \gamma_z \underbrace{\frac{s_{z,i}}{\mu_{z,i}} X_z}_{\substack{\text{Total variable cost of i} \\ \text{serving Domestic market}}} \nu_{z,i} + \sum_{j\in\Omega_{z^*}} \gamma_z \underbrace{\frac{s_{z^*,j}}{\mu_{z^*,j}} X_{z^*}}_{\substack{\text{Total variable cost of j} \\ \text{exporting to Foreign market}}} (1-\nu_{z^*,j}) + \underbrace{wF|\mathcal{D}_z|}_{\substack{\text{Fixed cost of} \\ \text{Dom. firms}}} \right] dz \quad (9)$$

with $\nu_{z,i}$ a dummy being one if firm i in z is from domestic origin, zero if firm i exports to market z. The first term in bracket corresponds to the share of total variable cost of domestic firms producing for the domestic market devoted to pay labor. The second one designates the analogous for domestic firms producing to export in the foreign market. Both production use domestic labor. The third term corresponds to the fixed cost the domestic firms pay to enter in their domestic market. Since there is no fixed cost to export nor import, this term is only paid once, in labor terms. In equation (9), one can see that a shock changing strategic interactions of firms in the foreign market can make exporting firms better off there, which affects their production since they sell more abroad, and which creates in general equilibrium wage spillovers in the domestic economy. It follows that, even a granular shock in the foreign economy that

impacted no domestic firm producing for domestic market, can actually change their production cost and therefore their productions.
Note that when writing (9) for the Foreign economy, the second term would sum over $\mathcal{A}_z \cap \mathcal{F}_z$, as Foreign firms can have an import ban. Economic sanctions can thus reshuffle the whole production costs inside of the target economy.

The market of goods and services clears in domestic trade and exports. Aggregate expenditure is equal to the sum of the aggregate revenue of the sector.

$$X_z = \int_0^1 \left[\sum_{i \in \mathcal{D}_k} \underbrace{\beta^z_{k,i} \frac{s_{k,i}}{\mu_{k,i}} X_k}_{\substack{\text{Revenue supplying} \\ \text{Domestic firms i} \in k}} \nu_{k,i} + \sum_{j \in \mathcal{A}_k(j) \cap \mathcal{F}_{k^*}} \underbrace{\beta^z_{k^*,j} \frac{s_{k^*,j}}{\mu_{k^*,j}} X_{k^*}}_{\substack{\text{Revenue supplying} \\ \text{Foreign firms } jin\text{k}^*}} (1 - \nu_{k^*,j}) \right] dk + \alpha_z I \tag{10}$$

with I the aggregate income of the domestic household and $\beta^z_{k,i} = \gamma^z_k \frac{P_z^{-\chi}}{P_z^{-\chi} + \rho^z_k P_{z^*}(i)^{-\chi}} = \gamma^z_k (b^z_{k,i})^{-\chi}$ the expenditure share of a domestic firm i in sector k to the domestic composite good z, and $\beta^z_{k^*,j}$ expressed similarly, the expenditure share of foreign firm j^* in sector k^* to the domestic composite good z. The first term in brackets corresponds to the aggregate revenue of supplying composite good z to firms in the domestic market from all possible sectors. The second term is similar but for firms in the foreign market. The term outside brackets corresponds to the final demand of domestic consumers.

As soon as an export control targets firm j^* in k^*, $\Omega_z(j^*) = \varnothing$ and since $\mathcal{A}_k(j^*) \cap \mathcal{F}_{k^*} \subset \Omega_z(j^*)$, the sanctioned firm is excluded from the second term in brackets. j^* needs to get input supply in its own economy. The export control component of sanctions impacts the foreign demand for composite goods and thus endogenously changes the domestic sectoral expenditures. Through that channel, export control backfires to the sanctioning economy.

The household income is equal, at the equilibrium, to its aggregate expenditure. The latter is composed of labor compensation, the aggregate profit of firms producing in the economy Π, and the aggregate trade balance D. This amounts to have:

$$I = wL + \Pi + D \tag{11}$$

where L is the labor force in the economy. The distinction with the literature on optimal trade policy is salient in equation (11) because there is no tariff revenue. The aggregate profit is composed of all the profits of domestic firms operating either on the domestic market or in the foreign one. It is distributed to the shareholders in the domestic economy and enters the income of the consumer.

$$\Pi = \int_0^1 \left[\sum_{i \in \Omega_z} \nu_{z,i} \pi_{z,i}(s_{z,i}) + \sum_{j \in \Omega_{z^*}} (1 - \nu^*_{z^*,j}) \pi^*_{z^*,j}(s^*_{z^*,j}) \right] dz \tag{12}$$

Since the model is static, I assume the aggregate trade deficit D exogenous such that $D + D^* = 0$. An economy can have endogenous trade deficit in some sectors, but has an exogenous trade balance in the aggregate. Concretely, sanctions can affect aggregate demand in some sectors since they are imperfectly substitutable, but cannot impact the aggregate resources available and thus shift the possible productivity frontier.

The general equilibrium is conditional on partial equilibrium. Equilibrium wages, expenditures, income and profits are given by equations (9), (10), (11) and (12). Aggregate price can be obtained *via* equation (2). Note that equations are almost symmetric for the foreign economy and that Walras' law allows to set the domestic wage as the numeraire in the model. This system of equations fully characterizes the general equilibrium of the model.

To find an equilibrium, I use the iterative procedure described in Appendix C. Starting from a guess of relative wages and expenditure in a monopolistic competition setup, I compute the partial equilibrium for every sector to find the allocation of a market between firms. I then use this information to compute aggregate quantities needed. Iterating over these nested equilibria provides the general equilibrium.

4.2 Comparative statics of the model

To study the mechanisms dictating microeconomic incidence and contributing to the macroeconomic incidence, I hold fixed the macroeconomic aggregates of the general equilibrium, and study a deviation from the partial equilibrium of one single sector, keeping all other sectors unaffected. It shows through which mechanisms firms adjust their price following a sanction.

Welfare decomposition. The welfare of an economy is measured by the real income of its consumers: $W = \frac{I}{P}$ with I the aggregate income defined in (11) of the economy and P the aggregate price index defined in (2).
Since the trade balance is exogenous ($dD = 0$) , labor force is constant ($dL = 0$), labor is inelastically supplied and domestic wage is the numeraire ($d \log wL = 0$), a change in welfare following a sanction is as follows:

$$d \log(W) = \frac{d\Pi}{I} - \int_0^1 \alpha_z d \log P_z dz \tag{13}$$

The two components in (13) correspond to the respective changes in producer surplus through

profits and consumer surplus through prices in general equilibrium. Note that a change in relative wage would be added for the symmetric version of (13) in the Foreign economy. This equation formalizes the two quantities that dictate macroeconomic incidence inside of the model. Two forces are impacting the real income: a change in worldwide profits, and a change in final price in the domestic economy.

Recalling that for a firm i in z, $\pi_{z,i} = (1 - \frac{1}{\mu_{z,i}})s_{z,i}X_z$ and that the price of the composite good varies following a shock $d\log P_z = \sum_{i\in\Omega_z} s_{z,i} d\log p_{z,i}$, it follows that three quantities in the model dictate the microeconomic incidence: market shares, markups and prices. I now study the change of those three quantities following the imposition of a sanction.

Impact of a sanction on the profit of the targeted firm. I can now relate a firm's sanction cost pass-through into its prices with its market share and import intensity. The starting point for this analysis is the optimal price setting equation (7), which can be log-differentiated to obtain: $d\log p_{z,i} = d\log \mu_{z,i} + d\log c_{z,i}$. Computing the two right-hand side terms unravels the mechanisms driving microeconomic incidence.
First, consider the markup term. Using (7) – (8) yields for a firm i in z:

$$d\log \mu_{z,i} = \Theta_{z,i}\, d\log s_{z,i} + \Xi_{z,i}\, d\log\left[(b^z_{z,i})^{-\chi}\right] \tag{14}$$

with $\Theta_{z,i} \equiv \frac{s_{z,i}(\sigma-1)\left[1-\gamma^z_z (b^z_{z,i})^{-\chi}\right]}{\mu_{z,i}\left(\epsilon_{z,i}-1\right)^2}$ the elasticity of the markup of firm i with respect to its own market share $s_{z,i}$ and $\Xi_{z,i} \equiv -\frac{s_{z,i}\gamma^z_z (b^z_{z,i})^{-\chi}}{\mu_{z,i}(\epsilon_{z,i}-1)}$ its elasticity with respect to its own import share reliance. The elasticity of the markup with respect to the market share is positive, which comforts the facts that large firms charge higher markups. In addition, appendix I shows that $\Theta_{z,i}$ is increasing in $s_{z,i}$ and decreasing in $(b^z_{z,i})^{-\chi}$. Intuitively, the higher the market share of a firm, the less elastic the demand it serves, and thus the higher markup it can charge. However, the higher the expenditure on its own domestic input, the higher the internalization of its own cost into its price, which in turn increases the elasticity of demand and thus makes firm i charge a lower markup. Remarking that $d\log[(b^z_{z,i})^{-\chi}] = -\chi\left(1-(b^z_{z,i})^{-\chi}\right)\left(d\log P_z - d\log P^*_z\right)$, one sees that terms of trade enters the change in markup of firm i in z following a sanction through the second term on the right hand side. Concretely, this additional term captures that the firms internalize that they are indirectly competing on the input market for the intermediate demand.

Combining equations (7) and (14), it appears that a sanction affects markups both directly and indirectly. The direct impact comes from the constrained nullity of the second term in (7), since a sanction forces $b^z_{z,i} = 1$. Once the supply chain is disrupted by the export control, the sanctioned firm must source domestically and face the internalization of its own price on its cost.

The indirect adjustment results from partial equilibrium movements in market shares. After the sanction enforcement, the unit cost of producing increases for the sanctioned firm, and does not for other firms in the sector. The targeted company becomes relatively less productive, and thus less competitive: it loses market shares. In turn, the sanctioned firm then faces a more elastic part of the demand, charges lower markup, and thus has lower profit.

Second, consider the cost term. The only endogenous term to the firm's program entering equation (5) is P_z. Denote $a_{z,i}$ the exogenous increase of cost induced by the export control. The change in producing cost of firm i in z following an export control is thus $d\log c_{z,i} = d\log a_{z,i} + [\gamma_z^z (b_{z,i}^z)^{-\chi}] d\log P_z$. Again, the elasticity of the unit cost with respect to the price of the composite good of its own sector depends on the reliance of the firm on the composite good from its own sector.

Impact of a sanction on the prices paid by consumer. I can reconstruct the variation of prices of all firms in the sector adding together those two pieces:

$$d\log p_{z,i} = \varrho_{z,i}\, d\log a_{z,i} + \psi_{z,i}\, d\log P_z \tag{15}$$

with $\varrho_{z,i} \equiv \frac{1}{1+(\sigma-1)\Theta_{z,i}}$ the pass-through of the increase of cost into final price, and $\psi_{z,i} \equiv \varrho_{z,i}\big[\gamma_z^z \big(b_{z,i}^z\big)^{-\chi} + (\sigma-1)\Theta_{z,i} - \chi\big(1 - \big(b_{z,i}^z\big)^{-\chi}\big)\Xi_{z,i}\big]$ the strategic complementarity of prices. Two results follow from this equation.

On the one hand, the pass-through of cost-increase to the final price ($\rho_{z,i}$) is decreasing in the market share ($s_{z,i}$). Firms with more market power want to keep serving an inelastic demand and thus decrease their markup to do so, thus passing through less. Since the markup elasticity to market share depends on sourcing strategy of firm i, the pass-through is increasing in the share of domestically sourced input $(b_{z,i}^z)^{-\chi}$. This follows from the fact that firms relying on their domestic input have lower markups, therefore they have less margin to absorb the shock and pass it more to final consumers.

On the other hand, appendix J shows that the strategic complementarity of prices is increasing in market shares. Intuitively, firms with large market share have lower prices than anyone. They can increase their price more when a shock shifts the pricing of every firm by the same share to the composite price. The variation is non-monotonic in $(b_{z,i}^z)^{-\chi}$.

Importantly, equation (15) demonstrates that an unsanctioned firm i in a sector z with sanctioned competitors, increases its price, even without incurring a shock (i.e. $d\log a_{z,i} = 0$). Intuitively, when a sanctioned firm increases its price, its competitors raise their price as well, thus shifting altogether to a new equilibrium with higher prices.

4.3 Sectoral vs. firm-level sanction in the model

Microeconomic incidence changes when shifting from firm-level to sectoral sanction. Equation (15) characterizes how a firm j^* in z^* adjusts its price following an economic sanction. One goal of this paper is to study how the microeconomic incidence varies when targeting one single firm or the whole sector. The difference between a firm-level and sectoral sanction is the number of targets. Summing over (15) I can show how much the pass-through of a firm j^* in z^* varies as other firms are also sanctioned at the same time.

Proposition 4.1. *Let $\mathcal{S}_{z^*}$ and $\mathcal{S}'_{z^*}$ be two sets of sanctioned firms included in $\mathcal{F}_{z^*}$. Assume that $\mathcal{S}_{z^*} \subsetneq \mathcal{S}'_{z^*}$ and $\forall j^* \in \mathcal{S}'_{z^*}$, $a_{z^*,j^*} = a$. Then for a firm $j^* \in \mathcal{S}_{z^*}$:*

$$\left.\frac{d\log p_{z^*,j^*}}{d\log a}\right|_{\mathcal{S}'_{z^*}} - \left.\frac{d\log p_{z^*,j^*}}{d\log a}\right|_{\mathcal{S}_{z^*}} = \frac{\psi_{z^*,j^*}}{1-\bar{\psi}_{z^*}} \sum_{j\in\mathcal{S}'_{z^*}\setminus\mathcal{S}_{z^*}} s_{z^*,j}\varrho_{z^*,j} \;>\; 0. \tag{16}$$

with $\bar{\psi}_{z^*} \equiv \sum_j s_{z^*,j}\,\psi_{z^*,j}$ The pass-through onto the final price of a given firm is strictly increasing in the total market share sanctioned in the sector.

Proposition 4.1 demonstrates that the more market shares are sanctioned inside of a sector, the higher the pass-through of every single sanctioned company. Since an additional sanctioned firm implies that more market shares in the sector are targeted, it directly follows that the pass through increases in the number of firms sanctioned. Concretely, when doing a firm-level sanction, the competition pressures the targeted company to pass through part of its increase of cost. As soon as more targets are named in the same sector, they all pass through more of the cost, since they all experience the same increase of cost. Doing so altogether, the sanctioned firms decrease their markup less, and thus decrease their profit loss following the sanction.

Since unsanctioned rivals also adjust, the final price of all firms in the sectors increase. It is unclear who pays more of the cost of the sanction between the sanctioned firm and the consumer.

Microeconomic incidence within the target economy. Combining and manipulating equations (13)–(15), I prove the second key theoretical result. Consider a firm-level sanction on $j^* \in \mathcal{F}_{z^*}$ raising its cost by $d\log a_{z^*,j^*} > 0$.

Define the damage inflicted to the targeted firm and on the representative household in the target economy,

$$D^{\pi^*}_{j^*} \equiv -\frac{d\pi^*_{z^*,j^*}}{d\log a_{z^*,j^*}}, \qquad D^{C^*}_{j^*} \equiv \alpha_{z^*} I^* \frac{d\log P_{z^*}}{d\log a_{z^*,j^*}}. \tag{17}$$

Proposition 4.2. *Define the precision of the damage when sanctioning firm j^* in sector z^*:*

$$\mathcal{R}_{z^*,j^*} \equiv \frac{D^{\Pi^*}_{j^*}}{D^{C^*}_{j^*}} = \mathcal{K}_{z^*} \times \frac{1}{\mu^*_{z^*,j^*}\,\varrho^*_{z^*,j^*}}, \qquad \mathcal{K}_{z^*} \equiv \left(1-\bar{\psi}_{z^*}\right)\left[1+\frac{\Psi_{z^*}}{\alpha_{z^*}I^*}\right], \tag{18}$$

with $\bar{\psi}_{z^*} = \sum_j s_{z^*,j}\psi_{z^*,j}$.

i) If $\sigma + s_{z^*,j^*}\left[(\sigma-1) + \gamma_{z^*}^{z^*}(b_{z^*,j^*}^{z^*})^{-\chi}\right] > 2$*, then* $\mathcal{R}_{z^*,j^*}$ *increases in* s_{z^*,j^*}.

ii) $\mathcal{R}_{z^*,j^*}$ *decreases in* $(b_{z^*,j^*}^{z^*})^{-\chi}$.

Proposition 4.2 splits the precision of a designation into a sectoral term and a firm-level term.

First, the sectoral term $\mathcal{K}_{z^*}$ depends on the aggregate quantities of composite good used as input, or for final consumption. The more of the composite good is bought by other firms rather than by households, the higher the precision. Intuitively, in an upstream sector, a given price increase lands on firms using the composite good as an input, and the part of it that reaches consumers is absorbed by the weighted strategic complementarity $\bar{\psi}_{z^*}$ in the sector. The weights follow from the fact that firms produce according to their market shares. All in all, policymakers should prefer sectors that sell mostly to other firms.

Second, the firm-level term $1/\left(\mu_{z^*,j^*}^* \varrho_{z^*,j^*}^*\right)$ collects two forces that both operate through the sanctioned firm size and that pull in opposite directions. The first component, $1/\mu_{z^*,j^*}^*$, measures the profit damage per unit of the target's revenue: it is the target's cost-to-revenue ratio. Larger firms charge higher markup, because they are more productive and thus have a smaller cost base. A given proportional cost raise decreases less their profits. This term decreases in s_{z^*,j^*}. The second term, $1/\varrho_{z^*,j^*}^*$, is the inverse of the consumer damage per unit of revenue. A larger sanctioned firm passes less of its cost increase into its final price, so the consumer price index moves by less. This term increases in s_{z^*,j^*}.

The estimations of σ revolves between 3 and 5, and the condition for monotonicity is satisfied as soon as $\sigma > 2$: it is thus always empirically satisfied. The share of the damage borne by the targeted firm relative to the foreign consumer rises with the size of the designated firm. A government that wishes to concentrate the harm on the aggregate firm profits rather than on the foreign households should therefore sanction the largest firms of a sector. Intuitively, proposition 4.2 shows that market share shields the sanctioned firm and the consumer in the target economy alike, but it shields the consumers faster.

Third, comparing equation (14) to proposition 4.2 illustrates a trade off for policymakers. On the one hand, enforcing an export control increases the magnitude of the shock to the targeted firm's cost and hits more its profit. On the other hand, the export control decreases the precision which implies that foreign consumers pay relatively more than the targeted firm. Targeting an export control to a firm with a very high reliance on imported inputs thus impacts more the target but disproportionately impacts the foreign consumer.

All these exercises were carried in partial equilibrium, holding fixed the prices of other sectors, wage and aggregate expenditure. Doing so cancels the spillover effects of sanctions and the substitution across sectors. I now turn to a simulation of the general equilibrium of the model to account for those forces.

4.4 Calibration

I now turn to a quantitative exercise to estimate welfare loss associated with different scenarios of sanctions. Using the features of my model, I can compare a sanction with only an import ban, a sanction with only a supply chain disruption, and both components at the same time.

The sectoral structure of my model follows the same Ricardian structure as in Gaubert and Itskhoki (2021)

Table 1: Parameter values from the literature

Parameter	Value	Source	Related to
σ	5	Broda and Weinstein (2006)	subst. elasticity btw. firms
$1+\chi$	2.4	Blaum et al. (2018)	subst. elasticity btw. input location
θ	4.4	Gaubert and Itskhoki (2021)	Distrib. of productivities
κ	1.3		Iceberg trade cost
μ_z	.095		Mean of distrib. of relative sectoral produ.
σ_z	1.4		Var of distrib. of relative sectoral produ.
$\forall z : \alpha_z$	54 values	My computation based on Timmer et al. (2015)	final consumption shares
$\forall z : \gamma_z$	54 values		labor share
$\forall (k,z) : \gamma_k^z$	(54 x 54) values		intermediate shares
$\forall (k,z) : \rho_k^z$	(54 x 54) values		import share

Notes: This table summarizes the parameter values used in the calibration.

Table 1 summarizes the calibration used in my simulation. I set the elasticity of substitution across final goods to be 5 following Broda and Weinstein (2006). The parameters μ_T and σ_T governing the relative comparative advantage of sectors in the economy, the shape parameter θ of the Pareto distribution of productivities and the iceberg trade cost κ are all based on the estimations of Gaubert and Itskhoki (2021). The parameters governing share spent by household on every sector α_z, share of labor γ_z used to produce, share of income from other sectors γ_z^k and share sourced abroad ρ_z^k are calibrated on the WIOD. The elasticity of substitution of import of inputs between domestic and foreign destination is calibrated based on Blaum et al. (2018). Both the estimation of Gaubert and Itskhoki (2021) and Blaum et al. (2018) rely on French firm-level data.

I then use WIOD to calibrate parameters related to sourcing decisions when considering France and other countries as Rest of the World, weighted by their trade share in French imports. I drop two sectors of services in the calibration. I further do bootstrap in finite sample to increase the number of sectors to match the assumption of the infinity of sectors. Details are provided in appendix O.

5 Quantitative results

The analytical results on comparative statics are carried out holding fixed the wage, aggregate expenditures and prices of sectors that are not sanctioned. However, the granularity of the model implies that even the most targeted sanction has an effect on its sectoral aggregate, thus also on other sectors through spillovers. Sanctions are thus not only reallocating market shares, they also modify the levels of production in every sector. The macroeconomic incidence consequently depends on the relative gains and losses in firm surplus. I thus calibrate the model and assess the welfare impact of sanctions on both economies.

As a baseline, I consider a free trade where every firm can engage freely in both imports and exports. I then compare this baseline to the enforcement of a fully prohibitive economic sanction against a whole sector and against the most productive firm in this same sector. The main results report sanctions against the most productive firm, based on the third stylized fact, which established that European sanctions target the large firms. Intuitively, it makes sense given they are the most likely to engage in import and export activities: the sanction has more grip. In the model, there is only one source of heterogeneity *ex ante* which uniquely identifies the target of the firm-level sanction.

The welfare is measured as real income in both the baseline and the sanction scenario. I report in the y-axis of the graphs in this section the relative change compared to the free trade baseline. Several sectoral characteristics could impact the aggregate welfare effects: the competition within the markets of the targeted firm(s), the magnitude of the trade flow severed, etc. To compare this sectoral dimension, I compute the sum of market shares of all foreign firms in the market of the sanctioning economy $\Lambda_z = \sum_{j \in \Omega_z} s_{z,j}(1 - \nu_{z,j})$. Taking the deciles of the distribution I can thus apply either a firm-level or sectoral sanction to all the sectors in the same bin. In practice I cannot sanction one single sector since sectors are infinitesimal in my framework. Appendix O provides more information on how I deal with the infinity of sectors. The graphs in this section thus present the deciles of foreign shares shocked on the x-axis. Intuitively, my measure proxies for import penetration: a sector in the first decile was not trading with the sanctioning economy *ex ante*, but a sector in the tenth decile has very productive foreign firms exporting to the sanctioning economy. In the latter, the sanction has the most grip. The gas sector in Russia for European sanctions is a good example of such a sector.

Since a fully prohibitive sanction encompasses the import ban and the export control, I then turn to studying each of them in turn, still comparing firm-level to sectoral sanctions. In practice doing so provides insight on the differences between partial and complete sanctions, according to the GSDB typology. The new prevalence of firm-level has been established studying complete sanctions since partial sanctions do not exist at the firm level.

Finally, to highlight the theoretical contribution of my model, I compare it to a benchmark

model ignoring the input-output structure of the economy, thus re-establishing the separability across market property. This allows me to study the extent of substitution of inputs and spillovers in the enforcement of sanctions.

5.1 What is the macroeconomic incidence of complete firm-level and sectoral sanctions?

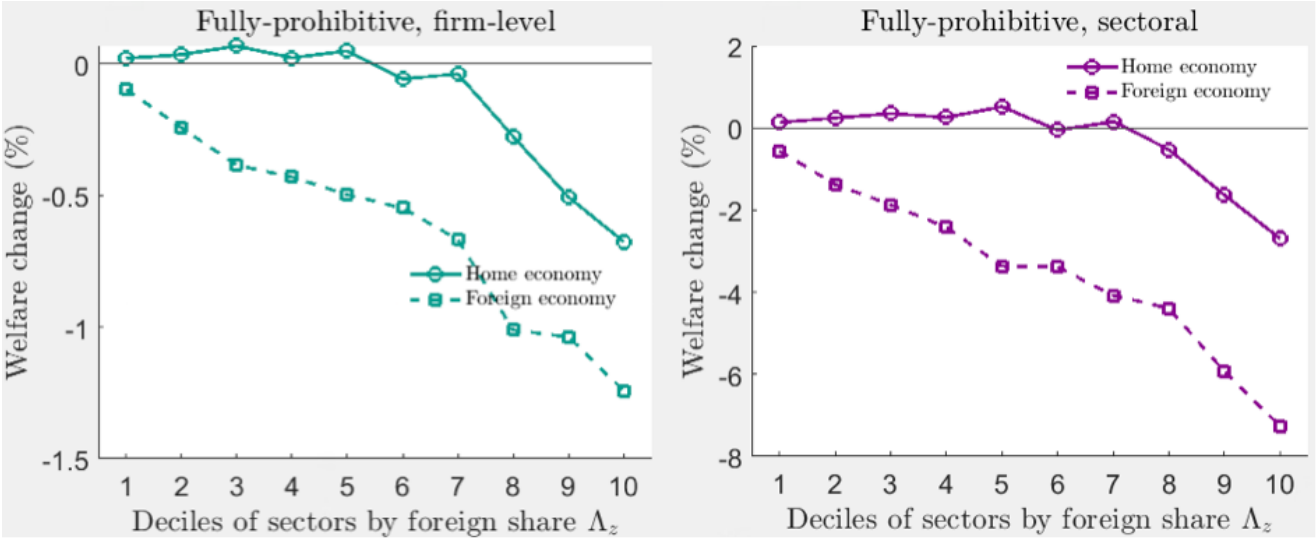
The macroeconomic incidence compares the aggregate welfare effect of the target economy, relative to the one of the sanctioning economy. Mathematically, this corresponds to dW^*/dW. The intuition is for the sanctioning economy to be better off than the target economy.

The quantitative simulation shows that a sanction upon the most productive firm of a sector imposes an average welfare loss of 0.14% for the sanctioning economy, and 0.62% for the target economy. Taking the ratio of the two, the macroeconomic incidence is 4.4. The target economy bears 4 times the cost of the sanctioning economy. Turning to a sectoral sanction upon all firms in the sector, the combination of import ban and export control reduces on average the welfare in the target economy sharply (3.5%), and in the sanctioning economy considerably less (0.3%). The macroeconomic incidence is thus 11.7. The target economy bears a burden 11 times higher than the sanctioning economy.

On average the loss of the sanctioning economy is of the same magnitude in both scenarios, which is surprising given the difference in magnitudes of the shock. On the contrary, the target economy is better off by a large amount when shifting from sectoral to firm-level sanctions. Comparing the asymmetries in the two scenarios, gives that a sectoral sanction has a welfare asymmetry 2.7 times larger than under firm-level sanctions. It follows that if the objectives of the policymaker were to damage the target economy relatively more than the sanctioning economy, they should, on average, prioritize sectoral sanctions.

The average macroeconomic incidence hides substantial heterogeneity across sectors yet. Figure 2 reveals three additional results.

Figure 2: Welfare Change Following a Fully Prohibitive Sanction Against a Single Firm or the Whole Sector



Notes: The graph at the left enforces a firm-level sanction on the most productive firm (i.e. with the highest $\varphi_{z,i}$) of the sectors in the decile of the x-axis. The graph at the right enforces a sanction against all firms in the sector at the same time. The y-axis represents the percentage change comparing the sanction enforcement to a free trade baseline. The welfare change is measured as a difference in real income between the two scenarios, in both economies.

First, an economic sanction always creates welfare losses in the target economy, but can create welfare gains in the sanctioning economy. It implies that the macroeconomic incidence can also be negative. Figure 2 further details that the welfare gains can be obtained predominantly in sectors with import penetration lower than the median. Comparing this result with equation (13) it must be that profit gains from domestic competitors of the targeted firm(s) outweigh the decrease in consumer surplus. Now turning to the deciles 8 to 10, it appears that the anti-competitive forces are overtaken by the loss in profit due to the loss of intermediate demand, and the loss in consumer surplus. Figure 2 thus illustrates a mechanism absent from the literature on optimal trade policy: the rent shifting of economic sanctions.

Second, the macroeconomic incidence is not constant, nor increasing linearly across sectors. It also depends on the foreign share, and thus the import penetration into the sanctioning economy. The macroeconomic incidence varies between -26 and 62 in this set of results summarized by table 2. The outlier values are driven by sanctions having an impact on the domestic economy close to zero. A negative incidence indicates that the sanctioning economy is gaining in welfare terms when the target economy is losing. A positive macroeconomic incidence indicates that both economies are losing. Computing the average of the macroeconomic incidence gives -1.35 for firm-level sanctions and 2 for sectoral sanctions. However the signs of these figures, even if it is impacted by outliers and denominators close to 0, reveal that the on average the firm-level sanctions seem to make the sanctioning economy better off, when the sectoral sanction is actually

creating losses on both sides.

Table 2: Macroeconomic Incidence (dW^*/dW) of a Fully Prohibitive Sanction By Import Penetration

Decile of Λ_z	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Firm-level	-4.8	-7.1	-5.8	-19.3	-10.4	9.4	16.9	3.7	2	1.8
Sectoral	-4.2	-5.6	-5.3	-9.3	-6.5	62.5	-26.2	8.4	3.6	2.7

Interestingly, in the first 5 deciles of the distribution, it appears that firm-level sanctions provide a higher macroeconomic incidence, making the sanctioning economy gain in welfare, and the target economy lose relatively more. In the 5 higher deciles, the sectoral sanction provides the highest macroeconomic incidence, in absolute terms. It shows that firm-level sanctions hit harder when making the sanctioning economy better off, and the sectoral sanctions hit harder when making the target economy worse off.

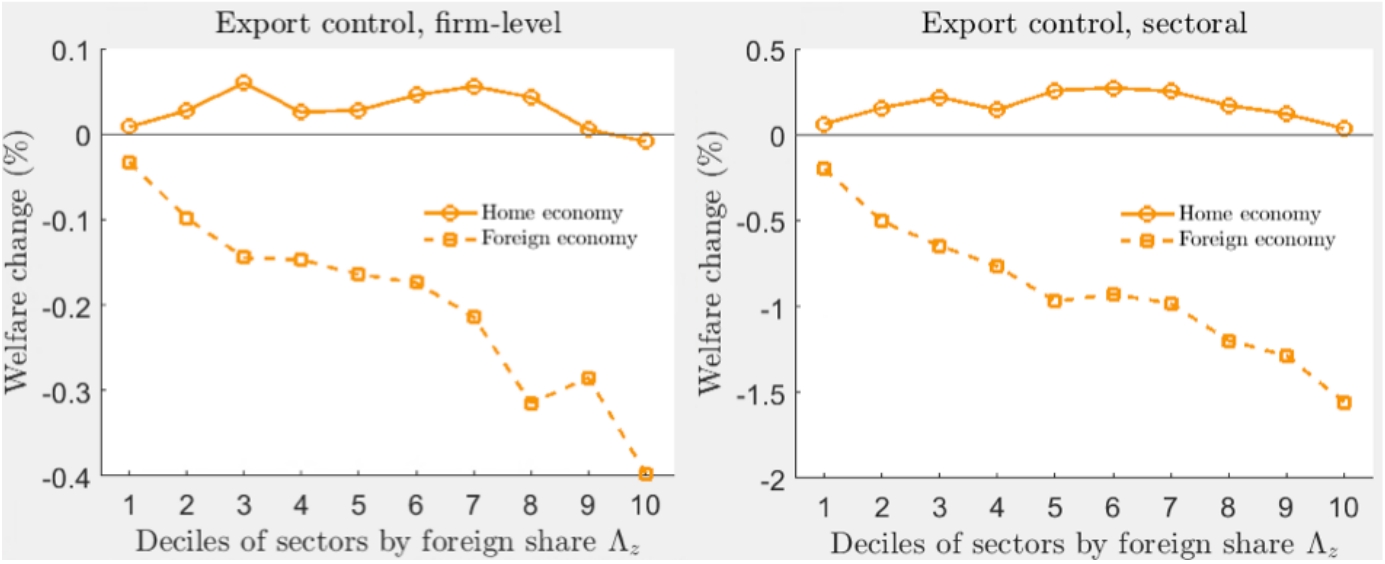
Third, I can compare the fourth stylized fact stating that European firm-level sanctions focus on the 5th to the 8th deciles, and sectoral sanctions hit the 4 highest deciles, to the graph 2 and table 2. In practice, the European Union, predominantly targets firm-level sanctions in the sectors where it gets welfare gains or small losses. Concretely these sectors coincide with those having a higher macroeconomic incidence. Turning to sectoral sanctions, it appears that the European Union targets predominantly the sectors in which the target economy incurs the highest costs but where the sanctioning economy bears its highest burden as well. In terms of macroeconomic incidence, those sectors are the ones with minimal macroeconomic incidence. All in all, the European Union enforces firm-level sanctions on the segment with most macroeconomic incidence, but targets its sectoral sanction on the segment with less macroeconomic incidence. Yet, because the losses of the target economy rise with import penetration, those sectoral embargoes are the ones that inflict the larger absolute loss.

However, the welfare gains in the sanctioning economy may come from anti-competitive forces in the market of the sanctioning economy, but also in the market of the target economy. To disentangle the two, I can enforce sanctions against the same target, but enforcing an import ban and an export control in turn.

5.2 In which market is the rent extraction strongest?

Impact of an export control. The export control can have an ambiguous impact on domestic firms as it cuts the target's international demand for domestic intermediate good, but can also advantage domestic exporters in the foreign market. It may also be that foreign firms in the foreign market get most of the market shares lost by the sanctioned firm, thus providing a negligible welfare effect. To test the magnitude of those two mechanisms, I thus enforce an export control

Figure 3: Welfare Change Following an Export Control Against a Single Firm or the Whole Sector



Notes: The graph at the left enforces a firm-level export control on the most productive firm (i.e. with the highest $\varphi_{z,i}$) of the sectors in the decile of the x-axis. The graph at the right enforces an export control against all firms in the sector at the same time. The y-axis represents the percentage change comparing the sanction enforcement to a free trade baseline. The welfare change is measured as a difference in real income between the two scenarios, in both economies.

against the most productive firm *ex ante*, and against the full sector. Figure 3 reveals two main results.

First, regardless of doing a firm-level or sectoral export control, the domestic economy experiences welfare gains no matter the import penetration of the sector targeted. The outlier in the 10th decile of the firm-level export ban is driven by a negligible welfare effect for the sanctioning economy. In the target economy, the welfare loss monotonically increases with the import penetration of the sector hit by the export control. The macroeconomic incidences are similar for firm-level and sectoral sanctions, except in the top two deciles, where the welfare change of the sanctioning economy approaches zero.

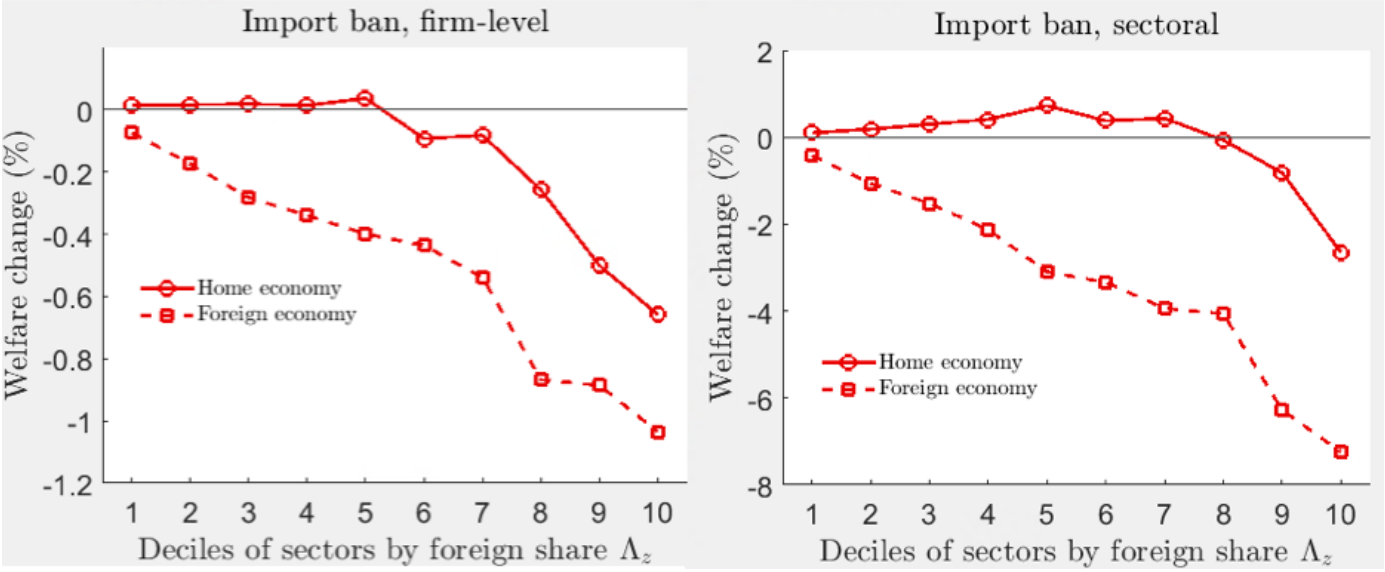
Second, the welfare of the domestic economy is increasing with an export control. Thus, the channel of the loss of intermediate demand decreasing the domestic profit is quantitatively dominated by the increase of exporting profits of exporting domestic firms.

Table 3: Macroeconomic Incidence (dW^*/dW) of an Export Control By Import Penetration

Decile of Λ_z	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Firm-level	-3.9	-3.6	-2.4	-5.8	-5.8	-3.8	-3.8	-7.3	-53.9	45.9
Sectoral	-3.2	-3.2	-3	-5.3	-3.8	-3.4	-3.9	-7	-10.6	-44.3

Impact of an import ban. The import ban can have an ambiguous impact on domestic welfare as it cuts the target off the domestic market, thus loosening competition and subtracting one

Figure 4: Welfare Change Following an Import Ban Against a Single Firm or the Whole Sector



Notes: The graph at the left enforces a firm-level import ban on the most productive firm (i.e. with the highest $\varphi_{z,i}$) of the sectors in the decile of the x-axis. The graph at the right enforces an import ban against all firms in the sector at the same time. The y-axis represents the percentage change comparing the sanction enforcement to a free trade baseline. The welfare change is measured as a difference in real income between the two scenarios, in both economies.

variety but can also advantage domestic firms in the domestic market. To test the magnitude of those two mechanisms, I thus enforce an import ban against the most productive firm *ex ante*, and against the full sector. Figure 4 reveals two main results.

First, comparing figures 4 and 2, the welfare changes following an import ban are very similar in magnitude and distribution across import penetration of the sector hit, to the one of a fully prohibitive sanction. It follows that most of the welfare impacts of a fully prohibitive sanction originate from the import ban component. When comparing the macroeconomic incidence in table 4, the export control seems to attenuate the incidences of the import ban. In level though, the export control seems to have little impact on the welfare loss of both economies, however it seems to drive most of the welfare gains of the domestic economy which are absent in a firm-level import ban. All in all, the export control has little impact on the level of welfare change, but is a clear driver of the welfare gains of the sanctioning economy.

Second, to get a sense of the addition of the supply chain in the model, I compare the impact of an import ban in my quantitative exercise to the case of an import ban in the model of Gaubert et al. (2021). Comparing the figure 4 with the appendix figure 17 shows that imposing the exact same shock without the input-output linkages drastically increases the magnitude of the impacts for both firm-level and sectoral sanctions. Additionally, without supply chain, economic sanctions can only be costly: the rent shifting mechanism arises through spillovers of economic sanctions on unsanctioned sectors through terms of trade appreciation.

Table 4: Macroeconomic Incidence (dW^*/dW) of an Import Ban By Import Penetration

Decile of Λ_z	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Firm-level	-4.7	-11	-15.2	-24.5	-10.9	4.7	6.6	3.4	1.8	1.6
Sectoral	-4.1	-5.6	-5	-5.2	-4.3	-8.8	-9	60.2	7.7	2.7

6 Conclusion

This paper studies who bears the cost of an economic sanction and shows that the answer depends on the scope of the embargo. I document a substitution from sectoral to firm-level sanctions that political objectives and measured success do not explain. I embed both sanction scopes in a quantitative model of granular firms sourcing inputs internationally. Two lessons emerge.

First, the sanction scope determines the microeconomic incidence within the target economy. Strategic interaction with unaffected rivals prevents a sanctioned firm from passing the cost on, so it absorbs the shock in its markup. Conversely, a sectoral embargo raises every cost together and is mostly paid by consumers. Second, the ranking of the two scopes reverses once the comparison moves to the macroeconomic incidence. A sectoral sanction inflicts on average eleven times more damage abroad than it imposes at home, against four times for a firm-level one, and close to six times more damage in total. The instrument policymakers are using relatively less is the more efficient geoeconomic tool in hurting the target economy. Their choice of scope cannot be rationalized by macroeconomic incidence, and is instead consistent with a preference defined over microeconomic incidence. Evaluations of sanctions that stop at aggregate national welfare therefore miss the margin on which policymakers appear to be choosing.

The results also bear on sanction policy. For the past decade, policymakers have been urged to target sanctions narrowly in order to spare civilian populations. The model supplies the economic foundation that this recommendation has lacked. A firm-level sanction spares the population not because it is gentler, but because oligopolistic competition makes the firm absorb the sanction shock. The same mechanism identifies which company to target in a firm-level sanction.

This paper defines the precision of a sanction as the loss of the targeted firm's profit relative to the loss borne by consumers in the target economy. A sanction is more precise when it targets i) the firms with the largest market shares in the sector, ii) sectors upstream in the supply chain. The model predicts a tradeoff when studying the reliance on imported inputs of the target firm. Targeting firms that import more makes a sanction more costly for the firm, but also makes the sanction less precise and thus borne by the foreign consumer. Integrating government objective functions optimizing on both macroeconomic and microeconomic incidences would turn these positive results into a theory of optimal sanction design, which is a natural avenue for future research.

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Robustness check: what if the biggest foreign firmswhere also sanctionned in the domestic economy? relax the falsematrix in Convergence.m

Robustness: bootstrap in finite sample: I create diagonal matrix by block (put 0 in anti-diagonale matrices). I could take a line and a row, put them at the end of the matrix and normalize as well (kind of a super wide economy that is less connected (bc de gammak are smaller).

A Sanction Trends from the Literature

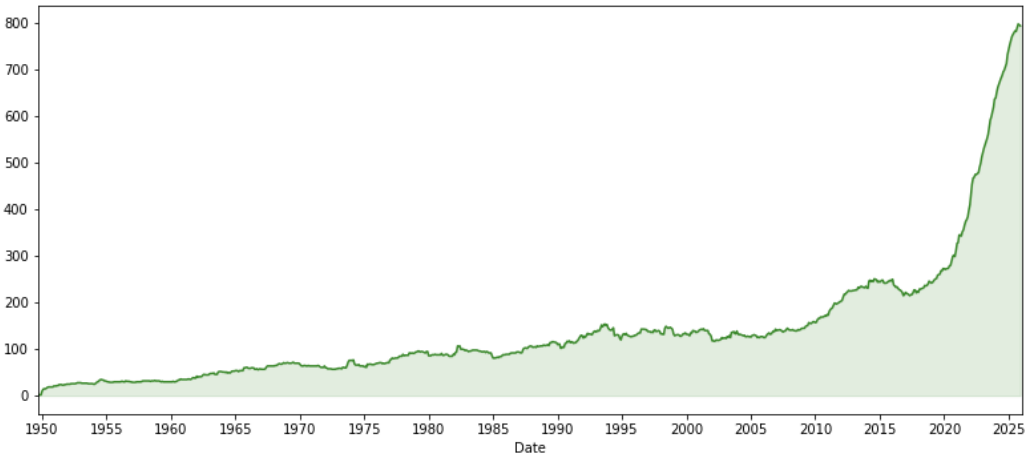
A.1 Literature review

Table 5: Table Comparing Theoretical Models of Granular Firms in Trade

Authors	**Year**	**Competition**	**Production technology**	**Equilibrium**
Grassi	2017	Oligopoly	Labour, domestic input	General
Amiti, Itskhoki, Koenings	2014, 2019	Oligopoly	Labour, input from different sources	Partial
Gaubert, Itskhoki, Vogler	2021	Oligopoly	Labour	General
di Giovanni, Levchencko, Méjean	2023	Monop. competition	Labour, input from different sources	General
This paper	–	Oligopoly	Labour, input from different sources	General

A.2 Aggregate facts about sanctions

Figure 5: Number of Active Sanction Cases Over Time



A.3 Travel bans

In the European sanction map, one can access the sanction cases imposed by the European Union (EU) since the creation of the Sanction Council in 2001. The European Union musts, in

Figure 6: Count of Countries Actively Targeted by a Sanction

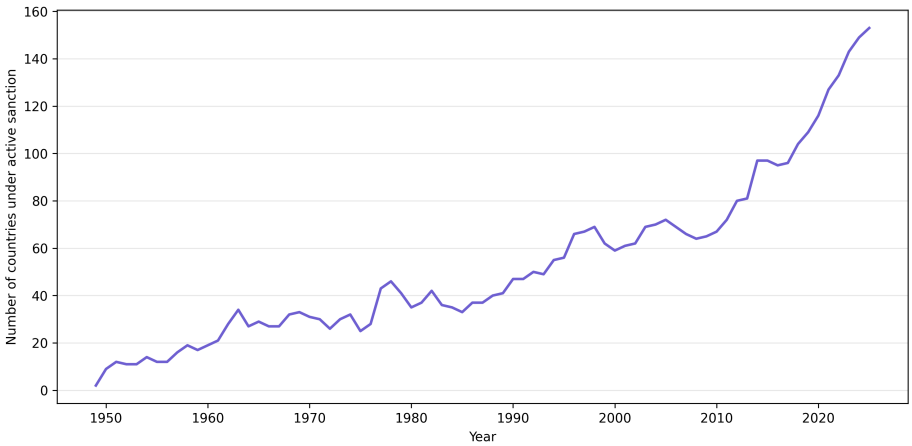
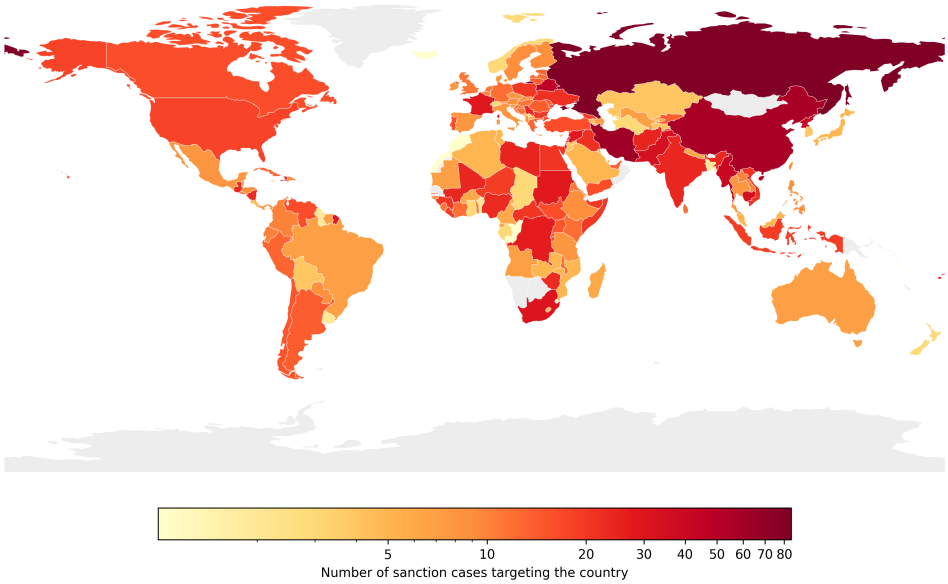


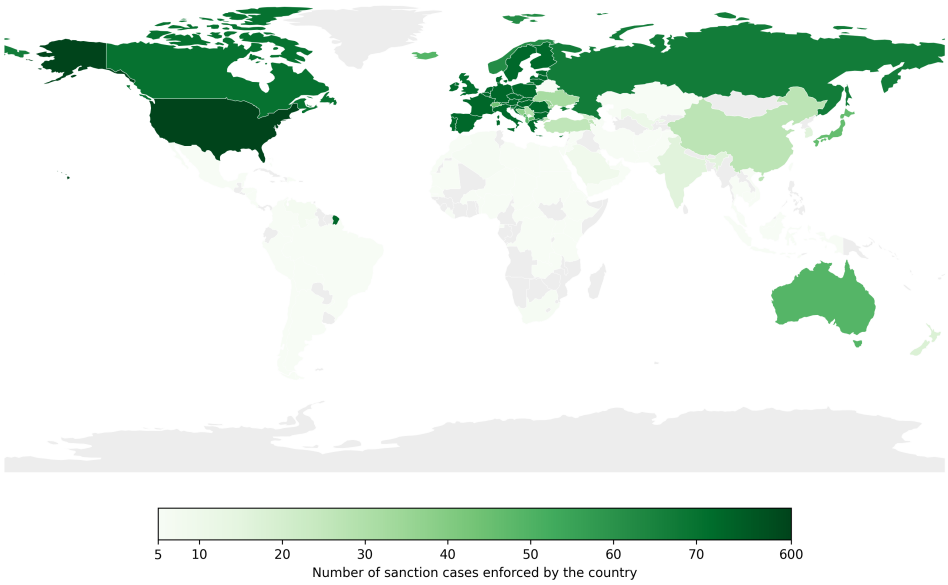
Figure 7: Map Counting Sanctions by Target Country



accordance with article X of GATT agreement, publish statements about every sanction, be it against a person or an entity. The EU also publishes reasons for the sanction. Every target can attack the decision of the EU to the EU court. The reasons for travel bans in some famous European programs is the sole relationship of the person to a sanctioned firm described in the FSDB (Montes, 2026). Even if the travel ban does not have a direct economic impact on the firm stated in the reason, it adds an indirect cost by constraining business trips of the workers in the firm.

The anecdotal evidence below was found by searching for keywords such as "head" and "director" in the EU sanction map. I only report the first result of this search.

Figure 8: Map Counting Sanctions by Sanctioning Country



Sanction case	**Listed person**	**Statement of reasons**
Russia	Leonid Kronidovich Ryzhenkin	Former Deputy general director for infrastructure projects at Stroigazmontazh (SGM) who since 2015 had supervised the construction of the bridge over the Kerch Strait (including the railway part of the bridge) connecting Russia and the illegally annexed Crimean peninsula.
Nicaragua	Juan Carlos Ortega Murillo	Director at Canal 8 and Difuso Comunicaciones. Leader of the 4th of May Sandinista Movement. He has contributed to restricting freedom of expression and freedom of the media.
Chemicals	Mohammed Khaled Nasri	Head of Institute 1000 of the SSRC, the division of the Scientific Studies and Research Centre (SSRC).
Terrorist Attacks	Omar Mahmood Irhayyim	Financial facilitator for Islamic State in Iraq and the Levant, listed as Al-Qaida in Iraq (QDe.115). Director of Al-Kawthar Money Exchange (QDe.157).
Myanmar	Naing Htut Aung	Director of International Group of Companies (IGGC) operating in the trading sector.
Moldova	Dumitru Chitoroagă	Director and administrator of a media trust which owns several media communication channels in the Republic of Moldova.

continued on next page

Sanction case	Listed person	Statement of reasons
Belarus	Igar Anatolievich Plysheuski	Managing Director of OOO Bergia Group, former Member of the Central Electoral Commission (CEC).
North Korea	Yu-ro Han	Director of Korea Ryongaksan General Trading Corporation.

B New Empirical Facts

Figure 9: Worlwide Enforcement of Sanctions Depending on the Scope of the Target

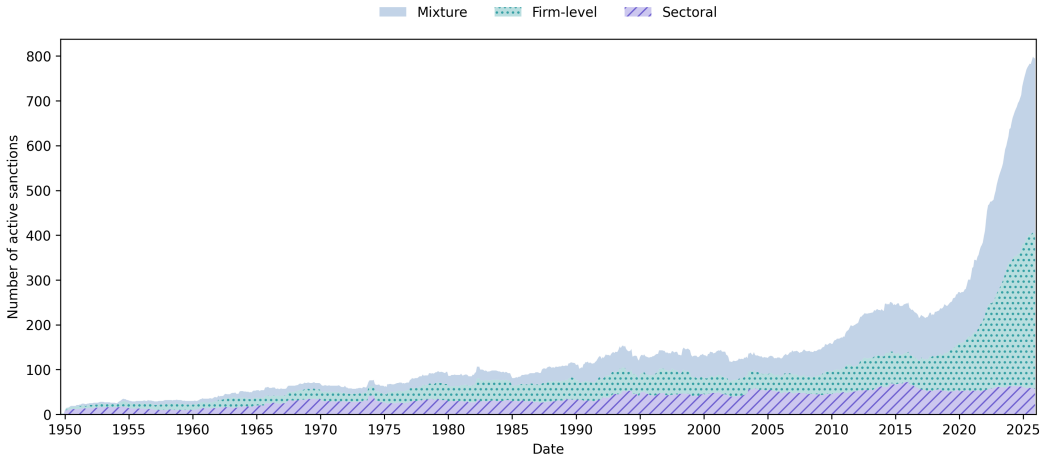
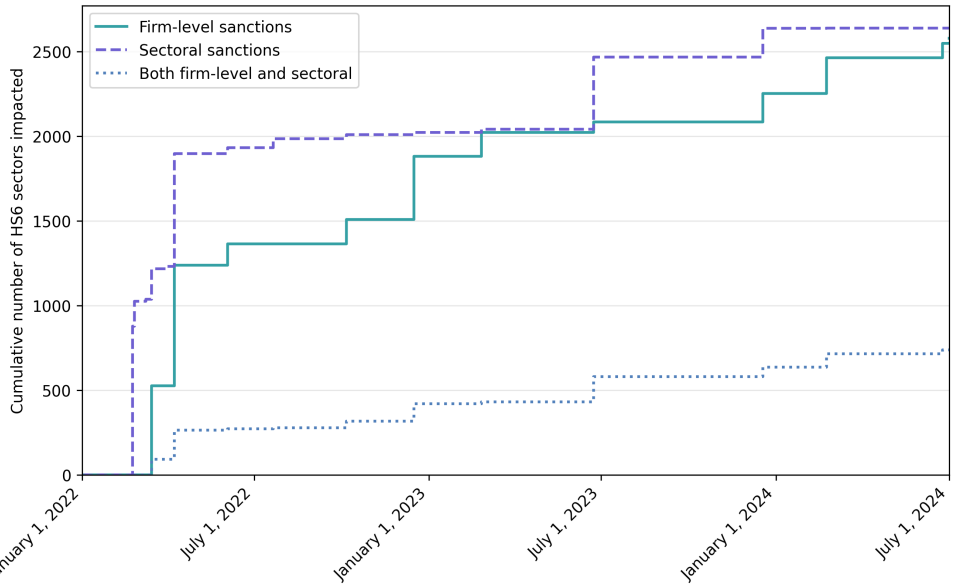
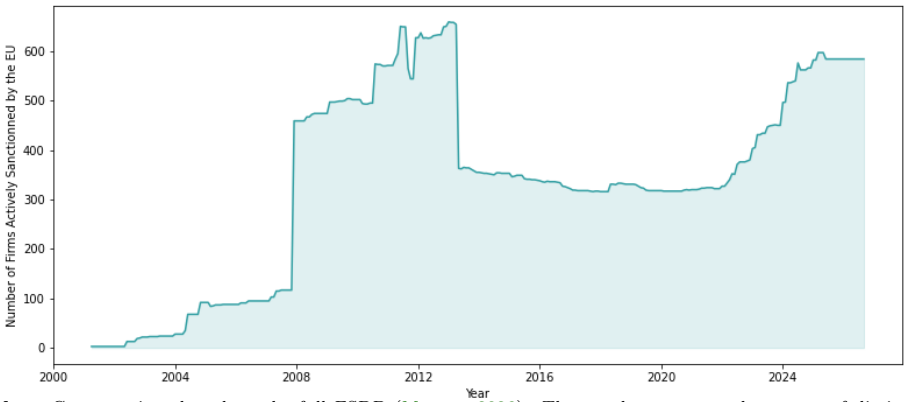


Figure 10: Count of HS6 Codes in Russia Sanctionned by the EU



Note: Sectoral sanctions are observed in Egorov et al. (2025), firm-level sanctions in Montes (2026). The graph shows the roll out of European sanctions against Russia. The discontinuous jumps originate from the different waves of sanctions enforced by the EU. The HS6 codes designated for both firm-level and sectoral sanctions originate from the assumption in Egorov et al. who do not distinguish complete ban against the full sectors, or bans against goods satisfying certain conditions.

Figure 11: Count of Firms Listed in Firm-Level Sanctions by the EU



Note: Computations based on the full FSDB (Montes, 2026). The graph represents the counts of distinctive firms across all 24 European programs of sanctions. The increase is not linear, as the Myanmar sanction episodes listed many firms in 2008, before lifting most sanctions in 2014 when a referendum was organized. The recent increase is mostly driven by the Russian program. This graph also shows that firms delisting is substantive.

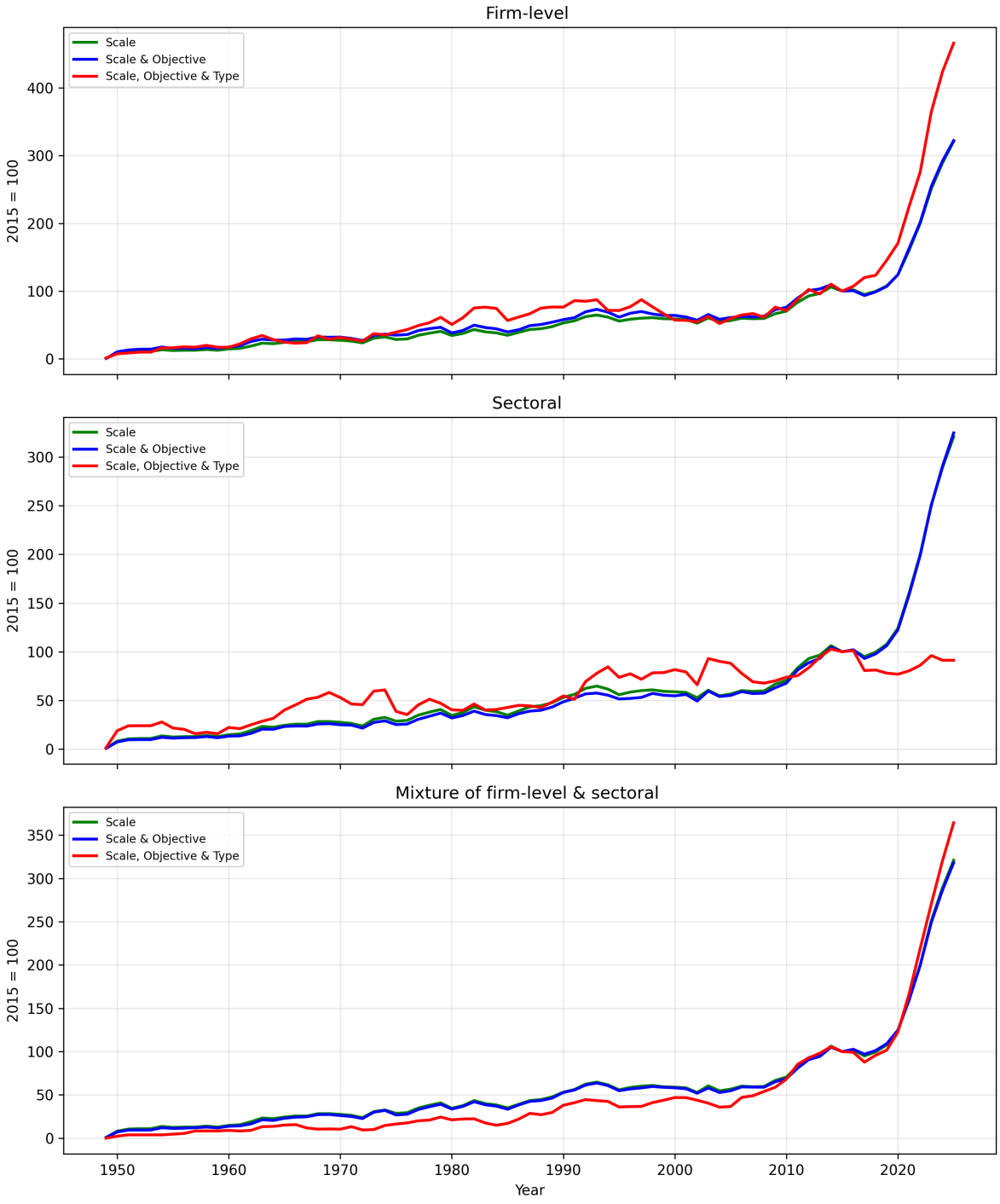
C Computational details

D Proofs

D.1 Derivation of the unit cost of a firm i in z producing a final good

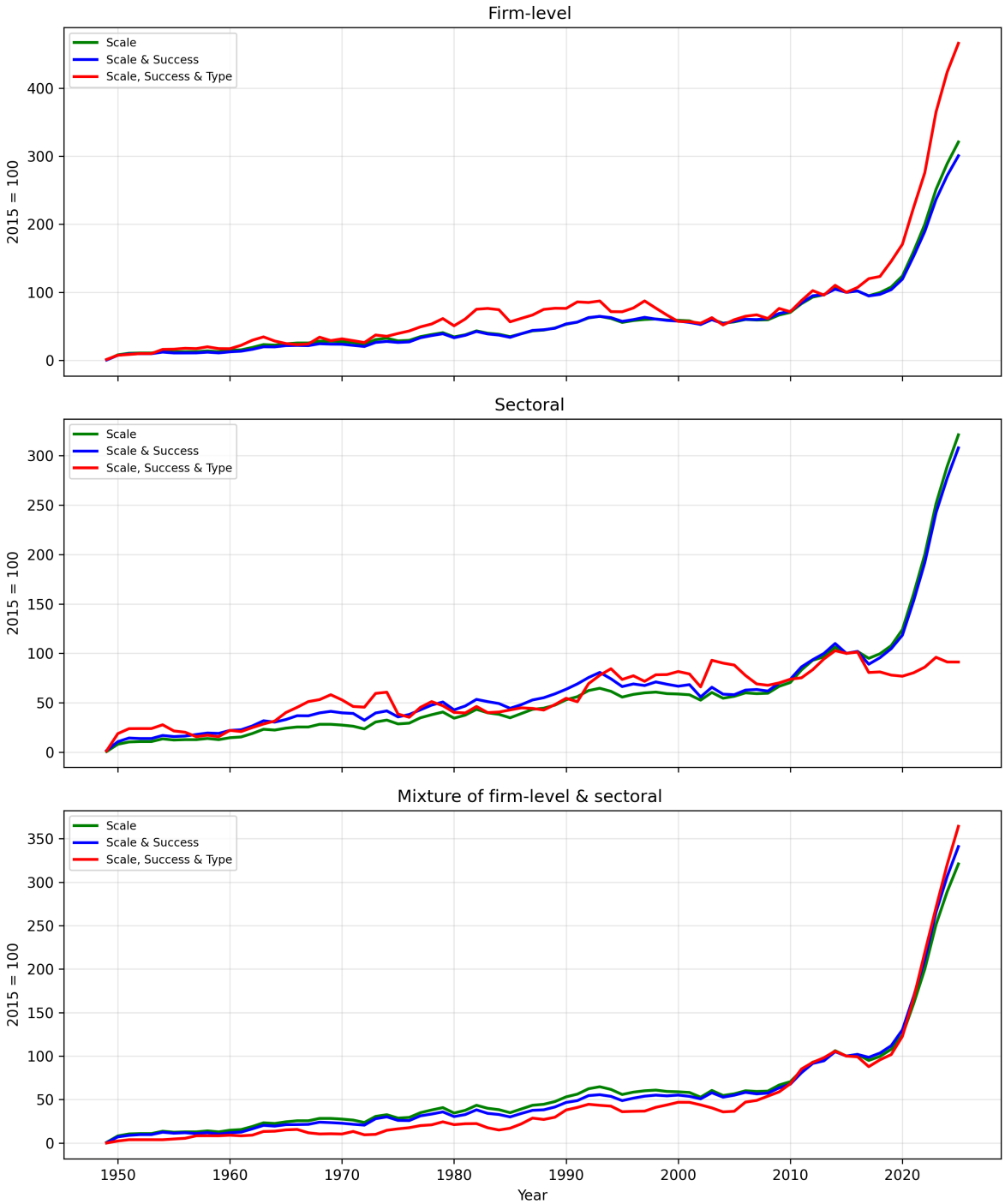
H1. The firm uses either labour ($l_{z,i}$) or intermediate input bundle ($M_{z,i}$). The production technology of the firm i in sector z is

Figure 12: Counterfactual Shift Share of Political Objective of Sanction Cases by Type Across Time



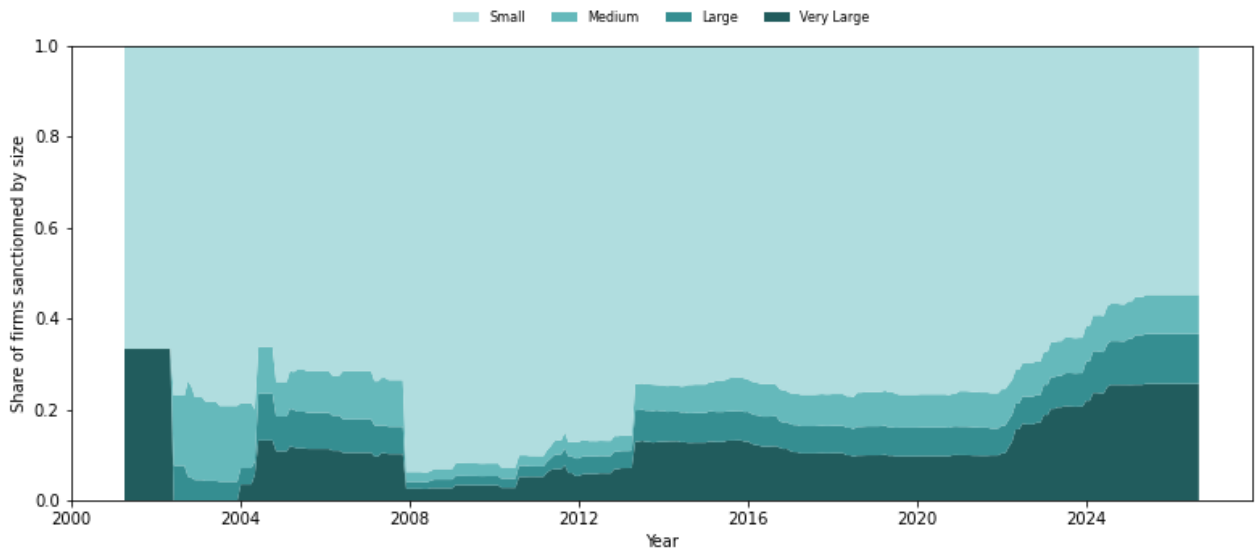
Notes: : This figure plots observed and counterfactual trends in evolutions of total number of active sanctions by type based on the statistical decomposition from equation (1). The green line plots the counterfactual level of sanctions by type with the same composition of political objectives and relative use of sanction types inside every objective in 2015. The blue line represents the level of sanctions by type, with the same use of sanction types inside every political objective as in 2015. The red line represents the actual observed number of sanctions by type, which consists of changes to the scale, composition of objectives, and type of sanctions relative to 2010 values. The top panel does the decomposition for sanction cases solely using firm-level sanctions. The middle panel does the same decomposition for sanctions using solely sectoral sanctions. The bottom panel studies cases of sanction mixing firm-level and sectoral sanctions.

Figure 13: Counterfactual Shift Share of Success Rate of Sanction Cases by Type Across Time



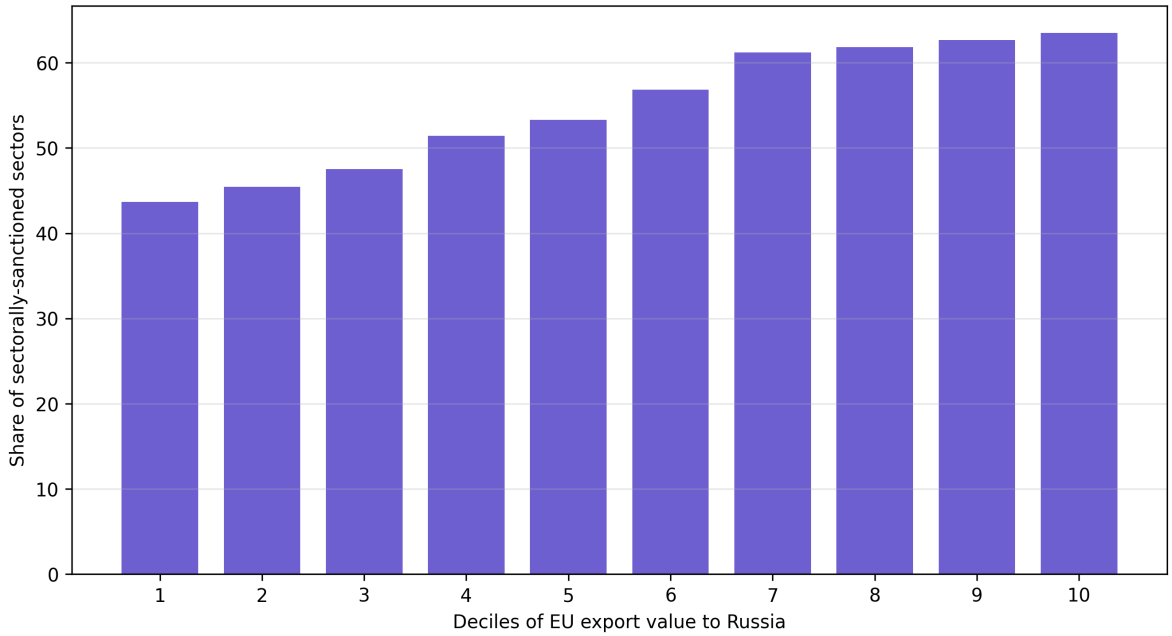
Notes: : This figure plots observed and counterfactual trends in evolutions of total number of active sanctions by type based on the statistical decomposition from equation (1). The green line plots the counterfactual level of sanctions by type with the same composition of success rate and relative use of sanction types inside every success rate in 2015. The blue line represents the level of sanctions by type, with the same use of sanction types inside every success rate as in 2015. The red line represents the actual observed number of sanctions by type, which consists of changes to the scale, composition of success rate, and type of sanctions relative to 2015 values. The top panel does the decomposition for sanction cases solely using firm-level sanctions. The middle panel does the same decomposition for sanctions using solely sectoral sanctions. The bottom panel studies cases of sanction mixing firm-level and sectoral sanctions.

Figure 14: Share of Firm Targeted by Size



Note: Computations based on FSDB (Montes, 2026). This graph shows the share of size of firms in active European sanctions across time. It shows that large firms have become more and more important in the set of European targets.

Figure 15: Share of Russian Sectors Sanctionned by Export Penetration of European Trade Flows into Russian Market

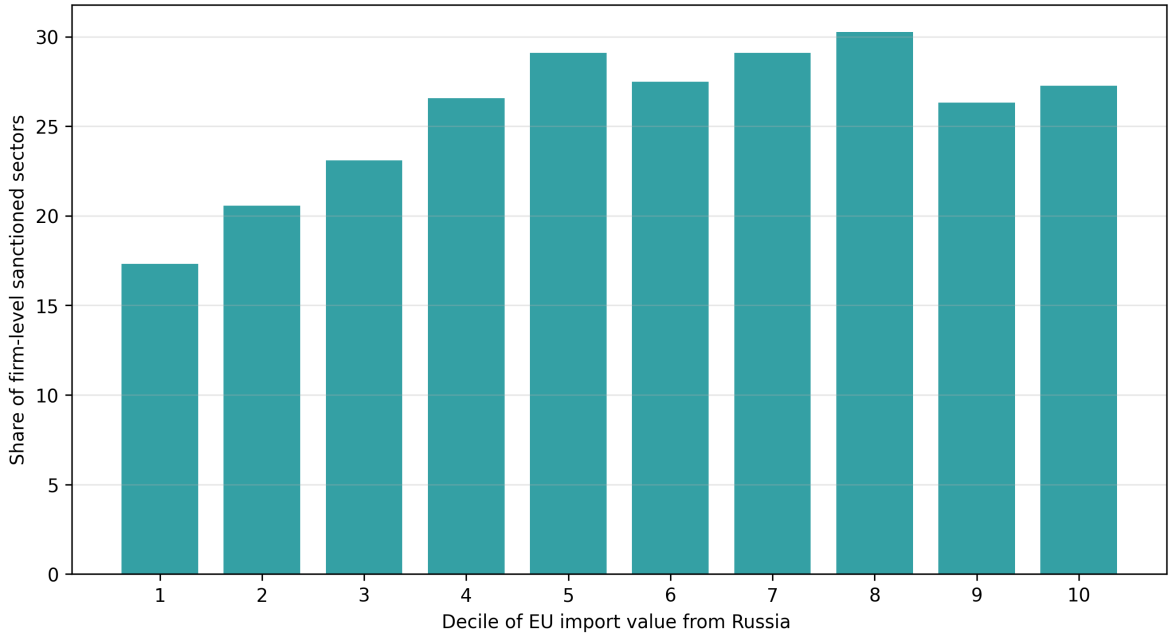


Note: Firm-level and sectoral sanctions are inferred from the typology of sanctions of the GSDB (Syropoulos et al., 2024). The graph highlights that firm-level sanction is the main driver of the novel increasing use of sanctions. A sanction encompassing firm-level and sectoral sanctions is counted as one in both curves.

$$q_{z,i} = \varphi_{z,i} l_{z,i}^{\gamma_z} M_{z,i}^{1-\gamma_z}$$

H2. The intermediate good bundle is made of a Cobb Douglass product of all sectors in the

Figure 16: Share of Russian Sectors Sanctionned by Import Penetration of Russian Trade Flows into European Market



Note: Firm-level and sectoral sanctions are inferred from the typology of sanctions of the GSDB (Syropoulos et al., 2024). The graph highlights that firm-level sanction is the main driver of the novel increasing use of sanctions. A sanction encompassing firm-level and sectoral sanctions is counted as one in both curves.

economy: $M_{z,i} = \exp\left[\int_0^1 \frac{\gamma_z^k}{1-\gamma_z} \log(M_{z,i}^{k,k^*})dk\right]$

H3. In every sector $k \in [0;1]$ input can be domestically sourced ($m_{z,i}^k$) or sourced abroad ($m_{z,i}^{k^*}$).

I follow Amiti, Itskhoki, Konings (2012), and aggregate them in a CES function. It follows:

$$\forall k \in [0;1] : M_{z,i}^{k,k^*} = [m_{z,i}^k{}^{\frac{\chi}{1+\chi}} + \rho_{z,i}^k{}^{\frac{1}{1+\chi}} m_{z,i}^{k^*}{}^{\frac{\chi}{1+\chi}}]^{\frac{1+\chi}{\chi}} \tag{19}$$

The lagrangian of the cost-minimizing problem of the firm is:

$$\begin{aligned}
\mathcal{L} = & wl_{z,i} + \int_0^1 P_k m_{z,i}^k dk + \int_0^1 (1+\tau_{z,i}^{k^*}) P_{k^*} m_{z,i}^{k^*} + wF) dk^* \\
& + \lambda_{z,i}\left[q_{z,i} - \varphi_{z,i} l_{z,i}^{\gamma_z} M_{z,i}^{1-\gamma_z}\right] \\
& + \Psi_{z,i}\left[M_{z,i} - \exp[\int_0^1 \frac{\gamma_z^k}{1-\gamma_z} \log(M_{z,i}^{k,k^*})dk]\right] \\
& + \delta_{z,i}\left[M_{z,i}^{k,k^*} - [m_{z,i}^k{}^{\frac{\chi}{1+\chi}} + \rho_z^k{}^{\frac{1}{1+\chi}} m_{z,i}^{k^*}{}^{\frac{\chi}{1+\chi}}]^{\frac{1+\chi}{\chi}}\right]
\end{aligned}$$

The FOC with respect to $l_{z,i}$ gives:

$$w = \lambda_{z,i}\gamma_z \frac{q_{z,i}}{l_{z,i}} \implies wl_{z,i} = \lambda_{z,i}\gamma_z q_{z,i} \tag{20}$$

The FOC with respect to $M_{z,i}$ gives:

$$\lambda_{z,i}(1-\gamma_z)\frac{q_{z,i}}{M_{z,i}} = \Psi_{z,i} \tag{21}$$

The FOC with respect to $M_{z,i}^{k,k^*}$ gives:

$$\delta_{z,i} = \Psi_{z,i}\frac{\gamma_z^k}{1-\gamma_z}\frac{M_{z,i}}{M_{z,i}^{k,k^*}} \tag{22}$$

The FOC with respect to $m_{z,i}^k$ gives:

$$P_k = \delta_{z,i}\left[\frac{M_{z,i}^{k,k^*}}{m_{z,i}^k}\right]^{\frac{1}{1+\chi}} \tag{23}$$

The FOC with respect to $m_{z,i}^{k^*}$ gives:

$$(1+\tau_{z,i}^{k^*})P_{k^*} = \delta_{z,i}\left[\rho_z^k\frac{M_{z,i}^{k,k^*}}{m_{z,i}^{k^*}}\right]^{\frac{1}{1+\chi}} \tag{24}$$

I inject (21) in (22):

$$P_k M_{z,i}^{k,k^*} = \lambda_{z,i}\gamma_z^k q_{z,i}\left[\frac{M_{z,i}^{k,k^*}}{m_{z,i}^k}\right]^{\frac{1}{1+\chi}}$$

I divide (24) by (23):

$$\left[\frac{(1+\tau_{z,i}^{k^*})P_{k^*}}{P_k}\right]^{1+\chi} = \rho_z^k\frac{m_{z,i}^k}{m_{z,i}^{k^*}} \tag{25}$$

I use (25) in (19):

$$\forall k\in[0;1] : M_{z,i}^{k,k^*} = m_{z,i}^k\left[1+\rho_z^k\left(\frac{(1+\tau_{z,i}^{k^*})P_k^*}{P_k}\right)^{-\chi}\right]^{\frac{1+\chi}{\chi}} \tag{26}$$

I inject this last equation in the right hand side of (D.1)

$$P_k M_{z,i}^{k,k^*} = \lambda_{z,i}\gamma_z^k q_{z,i}\left[1+\rho_z^k\left(\frac{(1+\tau_{z,i}^{k^*})P_k^*}{P_k}\right)^{-\chi}\right]^{\frac{1}{\chi}} \equiv \lambda_{z,i}\gamma_z^k q_{z,i} b_{z,i}^k \tag{27}$$

where I note $b_{z,i}^k = \left[1+\rho_z^k\left(\frac{(1+\tau_{z,i}^{k^*})P_k^*}{P_k}\right)^{-\chi}\right]^{\frac{1}{\chi}}$. It captures the enhancement of productivity allowed by a composite good k sourced abroad.

I inject (27) in (H2.) and obtain, because the first two terms do not depend on k:

$$M_{z,i} = \lambda_{z,i} q_{z,i} \exp\left[\int_0^1 \frac{\gamma_z^k}{1-\gamma_z} \log(\frac{\gamma_z^k b_{z,i}^k}{P_k}) dk\right] \tag{28}$$

I combine (28) and (20) to reexpress the production function

$$q_{z,i} = \varphi_{z,i} l_{z,i}^{\gamma_z} M_{z,i}^{1-\gamma_z} \implies \lambda_{z,i} = \frac{1}{\varphi_{z,i}} \times \frac{1}{\exp\left(\int_0^1 \gamma_z^k \log(b_{z,i}^k) dk\right)} \times \left[\frac{w}{\gamma_z}\right]^{\gamma_z} \exp\left(\int_0^1 \gamma_z^k \log(\frac{P_k}{\gamma_z^k}) dk\right) \tag{29}$$

If one injects (20), (26) and (27) in the original problem, it comes that $C_{z,i}(q_{z,i}) = \lambda_{z,i} q_{z,i}$

D.2 The aggregate quantity of the composite good is exogenous to a firm operating in z

This section demonstrates that the quantity of composite good in a sector z is, up to a $1/P_z$ term, exogenous to the program of a firm operating in this sector.

$$\begin{aligned}
M_z &= \int_0^1 \sum_{j \in \mathcal{A}_k \cap \mathcal{D}_k} m_{k,j}^z dk + \int_0^1 \sum_{j^* \in \mathcal{A}_k \cap \mathcal{F}_{k^*}} m_{k^*,j^*}^z dk^* \\
&\overset{(26)}{=} \int_0^1 \sum_{j \in \mathcal{A}_k \cap \mathcal{D}_k} \frac{M_{k,j}^{z,z^*}}{b_{k,j}^z} dk + \int_0^1 \sum_{j^* \in \mathcal{A}_k \cap \mathcal{F}_{k^*}} \frac{M_{k^*,j^*}^{z^*,z}}{b_{k^*,j^*}^z} dk^* \\
&\overset{(27)+sym}{=} \int_0^1 \sum_{j \in \mathcal{A}_k \cap \mathcal{D}_k} \frac{c_{k,j} \gamma_k^z q_{k,j}}{P_z} dk + \int_0^1 \sum_{j^* \in \mathcal{A}_k \cap \mathcal{F}_{k^*}} \frac{c_{k^*,j^*} \gamma_k^z q_{k^*,j^*}}{P_{z^*}} dk^* \\
&\overset{\times \frac{P_z}{P_z} \text{ RHS}}{=} \frac{1}{P_z} \left[\int_0^1 \sum_{j \in \mathcal{A}_k \cap \mathcal{D}_k} c_{k,j} \gamma_k^z q_{k,j} dk + \int_0^1 \sum_{j^* \in \mathcal{A}_k \cap \mathcal{F}_{k^*}} \frac{P_z}{P_{z^*}} c_{k^*,j^*} \gamma_k^z q_{k^*,j^*} dk^*\right] \\
&= \frac{1}{P_z} \left[\int_0^1 \sum_{j \in \mathcal{A}_k \cap \mathcal{D}_k} c_{k,j} \gamma_k^z q_{k,j} dk + \int_0^1 \sum_{j^* \in \mathcal{A}_k \cap \mathcal{F}_{k^*}} \left((b_{k^*,j^*}^z)^\chi - 1\right)^{\frac{-1}{\chi}} c_{k^*,j^*} \gamma_k^z q_{k^*,j^*} dk^*\right]
\end{aligned}$$

where the passage to the last line comes from the definition of b_{k^*,j^*}. The first term corresponds to the total variable cost of the domestic economy. Given that all firms in Domestic are infinitesimal to the whole economy, they can not internalize their impact on that first term. The second term corresponds to the total variable cost for firms in the foreign economy, but since the domestic firms can not have affiliate abroad, the firm has no impact on it.

$1 + \Psi^*_{z^*}/\left(\alpha_{z^*} I^*\right)$ exceeds one by the sector's intermediate-demand intensity: the more upstream the sector, the smaller the fraction of a price increase that lands on households, and the higher the precision. The factor $1 - \bar{\psi}^*_{z^*}$ damps precision in sectors with strong strategic complementarity, where the initial price increase is amplified through the roundabout loop before reaching consumers. Both are observable in input–output data.

O Infinity of sectors - bootstrap in finite sample

P Simulation

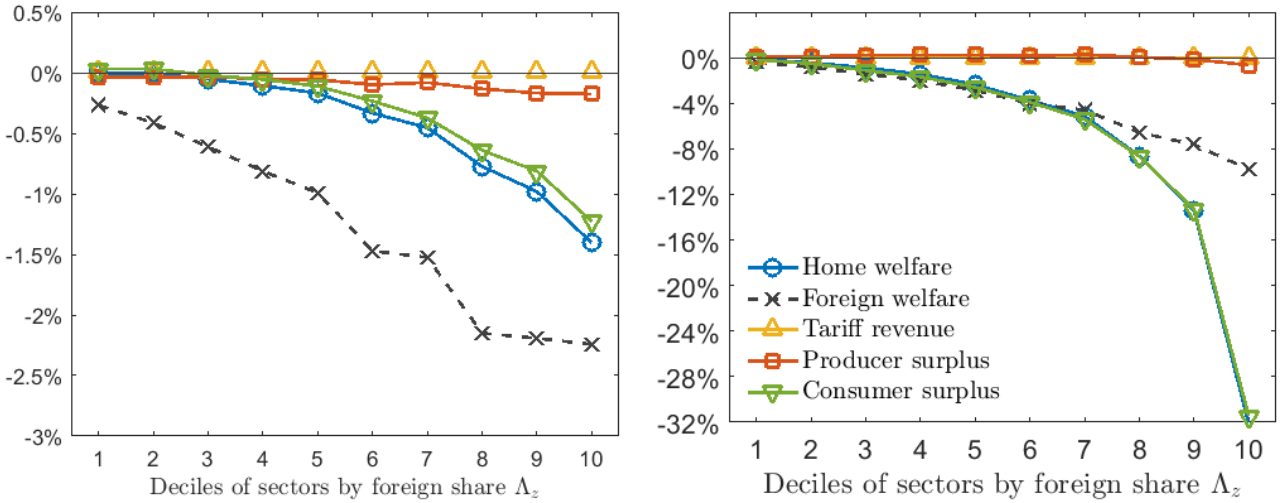
P.1 Simulation - Iterative algorithm

I take as a first guess for markups the constant case $\frac{\sigma}{\sigma-1}$. I also take for initial guess of marginal cost, the one that is given in the GIV model, ie. with just labour : $\frac{w}{\varphi}$. I then only keep the firms which productivity would make them active in the market for constant markups. I do not want to study shadow firms.

For the GE, I take as an initial guess for sectoral expenditure, the initial guess of GIV where I multiply for each sector by the share of final expenditure in that sector. I create the coefficient if the model only had final demand.

P.2 Simulation - Additional results

Figure 17: Welfare Decomposition of the Impact of the import ban component of sanctions.



(B) Firm-level Sanction (A) Sectoral Sanction

Notes: Decomposition of the welfare effect $\widehat{W}_z$ at home into two components according to equation (13), and the welfare effect abroad, from a sectoral and a firm sanction against the most productive firm of the sector, by deciles of sectors split by foreign comparative advantage. The y-axis represents the percentage change compared to a benchmark of free trade.

Q Robustness

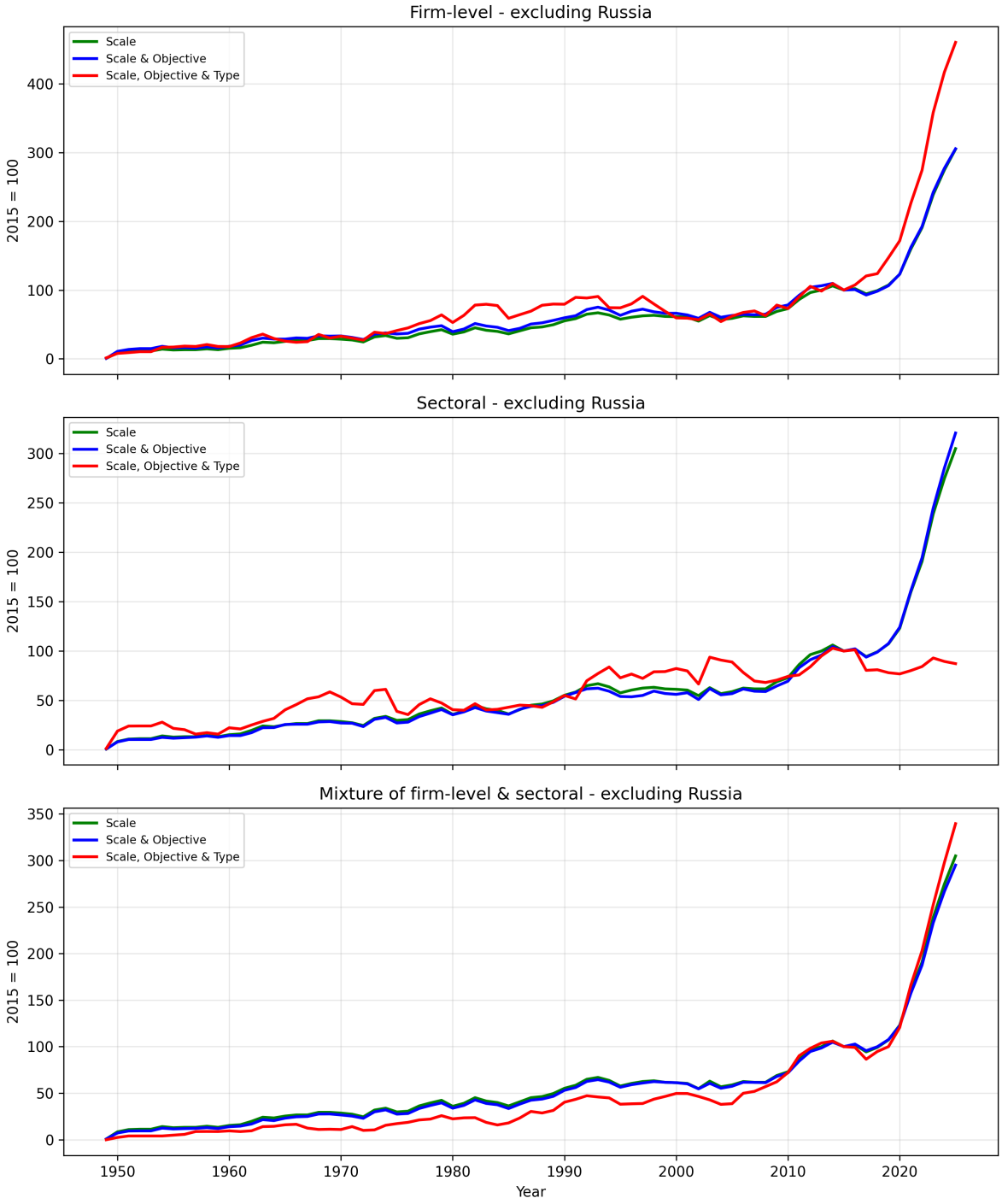
Q.1 Robustness - statistical decomposition

Q.1.1 Geopolitical criterion: political objective

Q.1.2 Geopolitical criterion: success rate

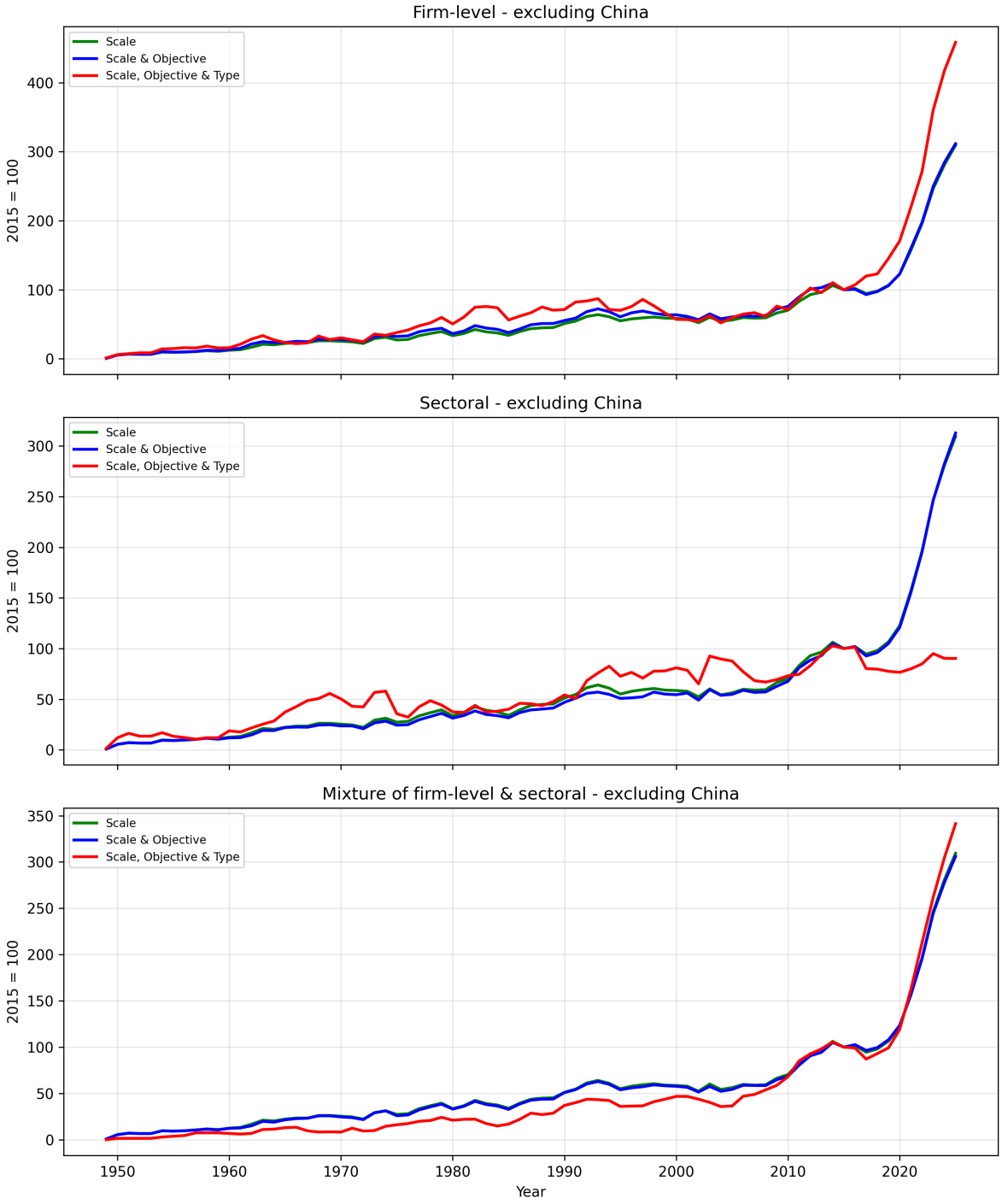
Q.2 Robustness - parameters in the model

Figure 18: Counterfactual Shift Share of Political Objective of Sanction Cases by Type Across Time, Excluding a Major Contemporaneous Target



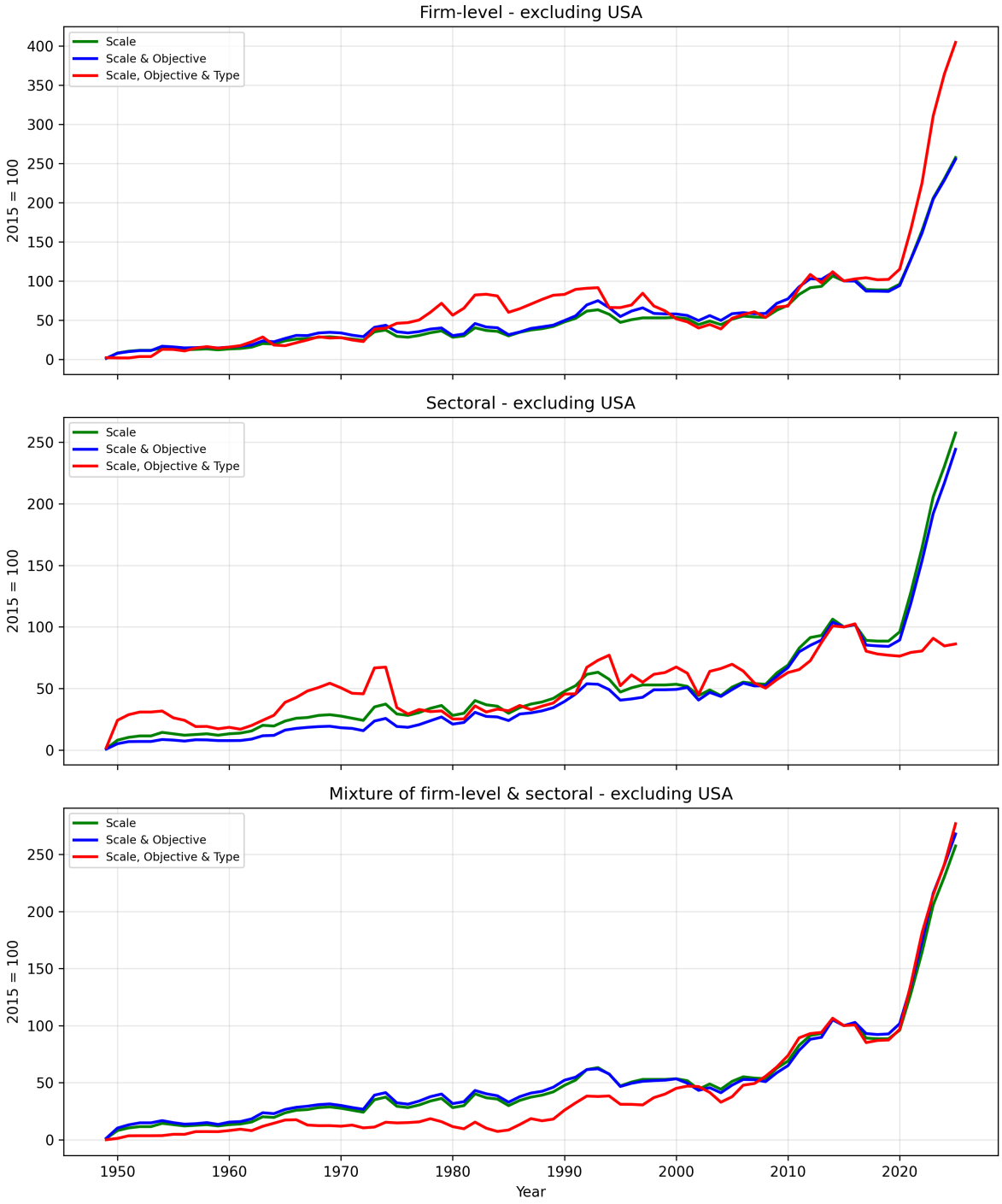
Notes: : This figure plots observed and counterfactual trends in evolutions of total number of active sanctions by type based on the statistical decomposition from equation (1). The green line plots the counterfactual level of sanctions by type with the same composition of political objectives and relative use of sanction types inside every objective in 2015. The blue line represents the level of sanctions by type, with the same use of sanction types inside every political objective as in 2015. The red line represents the actual observed number of sanctions by type, which consists of changes to the scale, composition of objectives, and type of sanctions relative to 2010 values. The top panel does the decomposition for sanction cases solely using firm-level sanctions. The middle panel does the same decomposition for sanctions using solely sectoral sanctions. The bottom panel studies cases of sanction mixing firm-level and sectoral sanctions.
Excluding Russia as a target from the GSDB drops 75 of 1794 cases. The y-axis is normalized to the 2015 levels and should not be read as a count level.

Figure 19: Counterfactual Shift Share of Political Objective of Sanction Cases by Type Across Time, Excluding a Major Contemporaneous Target



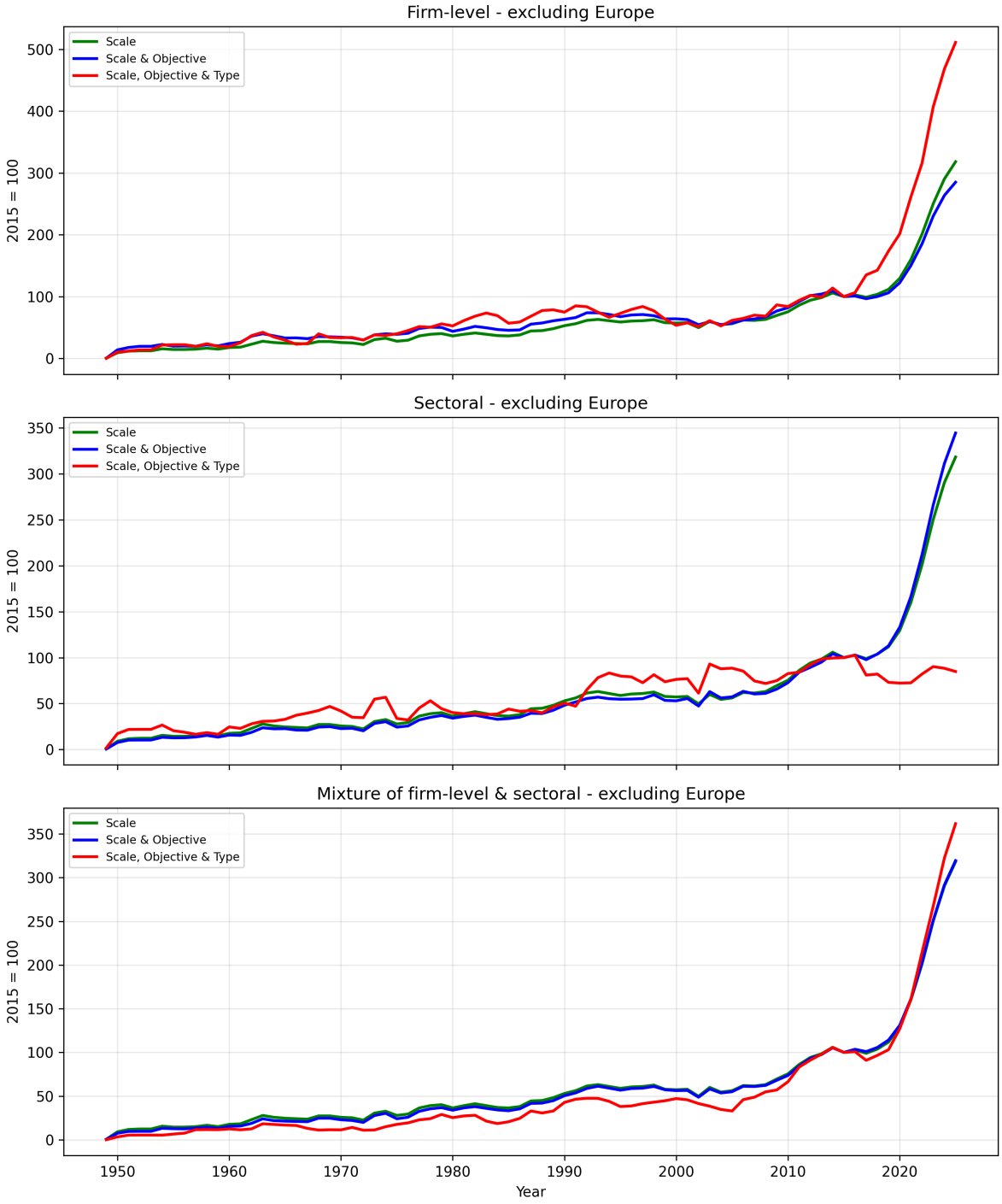
Notes: : This figure plots observed and counterfactual trends in evolutions of total number of active sanctions by type based on the statistical decomposition from equation (1). The green line plots the counterfactual level of sanctions by type with the same composition of political objectives and relative use of sanction types inside every objective in 2015. The blue line represents the level of sanctions by type, with the same use of sanction types inside every political objective as in 2015. The red line represents the actual observed number of sanctions by type, which consists of changes to the scale, composition of objectives, and type of sanctions relative to 2010 values. The top panel does the decomposition for sanction cases solely using firm-level sanctions. The middle panel does the same decomposition for sanctions using solely sectoral sanctions. The bottom panel studies cases of sanction mixing firm-level and sectoral sanctions.
Excluding China as a target from the GSDB drops 57 of 1794 cases. The y-axis is normalized to the 2015 levels and should not be read as a count level.

Figure 20: Counterfactual Shift Share of Political Objective of Sanction Cases by Type Across Time, Excluding a Major Sanctioning Economy



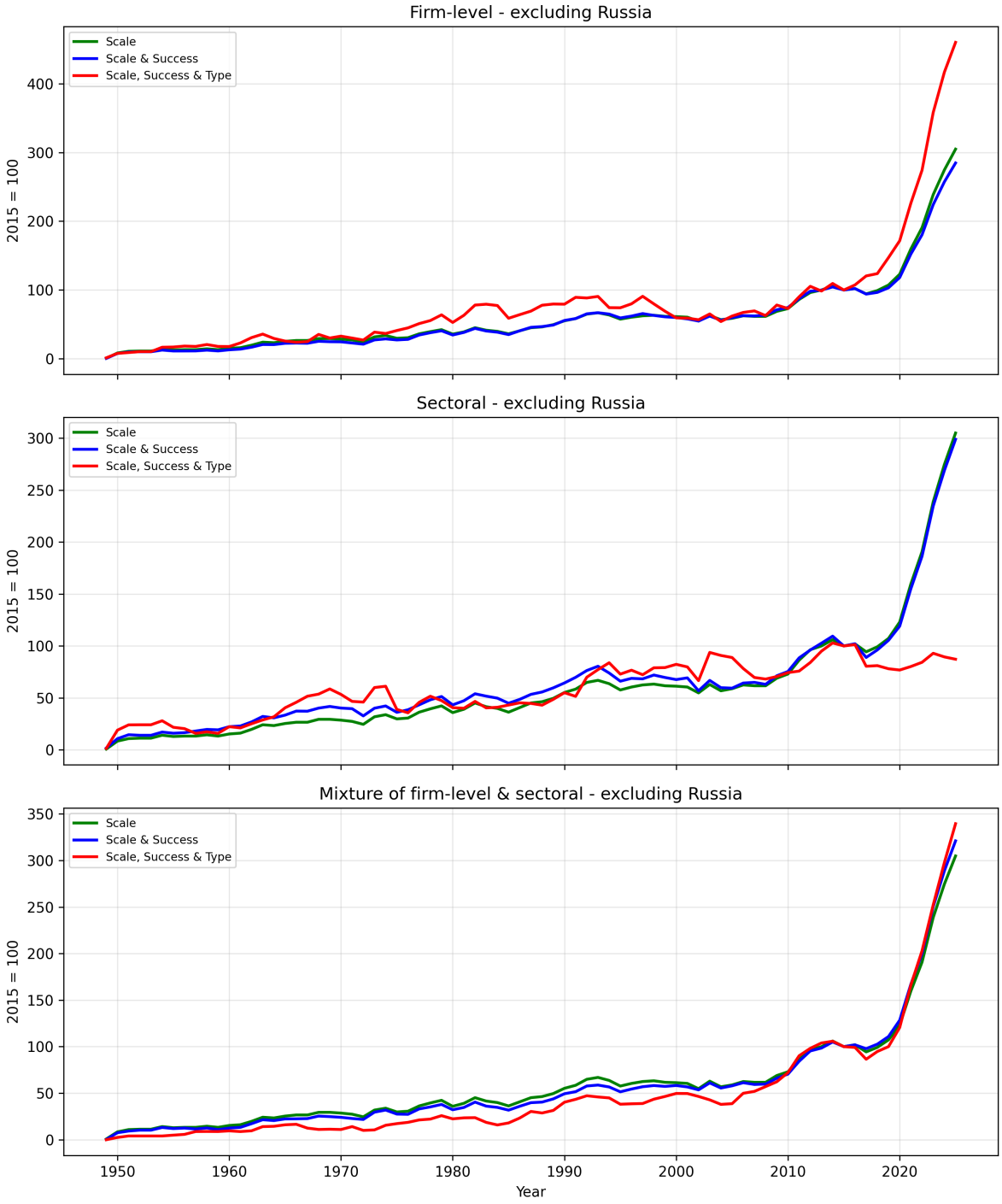
Notes: : This figure plots observed and counterfactual trends in evolutions of total number of active sanctions by type based on the statistical decomposition from equation (1). The green line plots the counterfactual level of sanctions by type with the same composition of political objectives and relative use of sanction types inside every objective in 2015. The blue line represents the level of sanctions by type, with the same use of sanction types inside every political objective as in 2015. The red line represents the actual observed number of sanctions by type, which consists of changes to the scale, composition of objectives, and type of sanctions relative to 2010 values. The top panel does the decomposition for sanction cases solely using firm-level sanctions. The middle panel does the same decomposition for sanctions using solely sectoral sanctions. The bottom panel studies cases of sanction mixing firm-level and sectoral sanctions.
Excluding the USA from the sanctioning economies in the GSDB drops 686 of 1794 cases. The y-axis is normalized to the 2015 levels and should not be read as a count level.

Figure 21: Counterfactual Shift Share of Political Objective of Sanction Cases by Type Across Time, Excluding a Major Sanctioning Economy



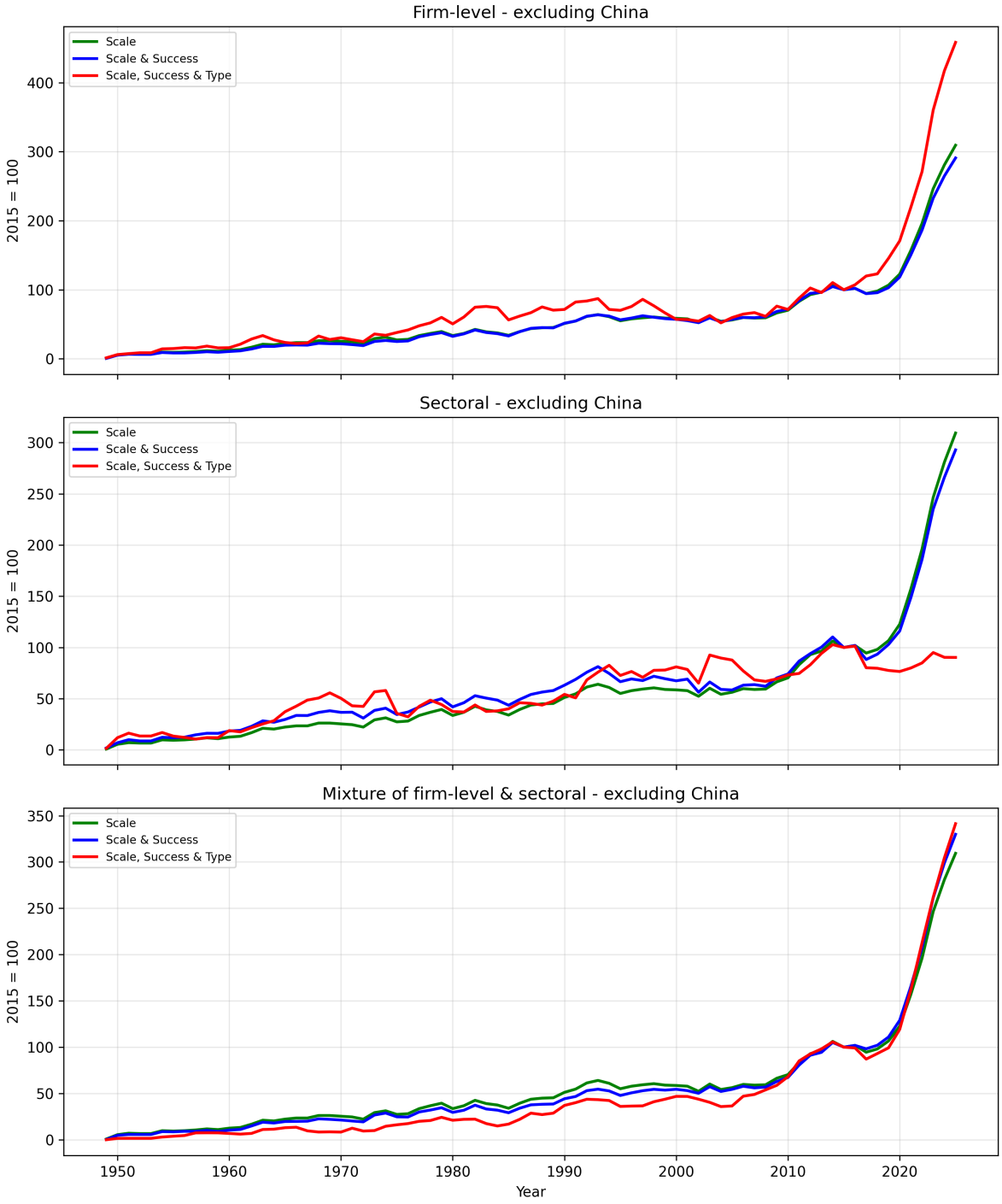
Notes: : This figure plots observed and counterfactual trends in evolutions of total number of active sanctions by type based on the statistical decomposition from equation (1). The green line plots the counterfactual level of sanctions by type with the same composition of political objectives and relative use of sanction types inside every objective in 2015. The blue line represents the level of sanctions by type, with the same use of sanction types inside every political objective as in 2015. The red line represents the actual observed number of sanctions by type, which consists of changes to the scale, composition of objectives, and type of sanctions relative to 2010 values. The top panel does the decomposition for sanction cases solely using firm-level sanctions. The middle panel does the same decomposition for sanctions using solely sectoral sanctions. The bottom panel studies cases of sanction mixing firm-level and sectoral sanctions.
Excluding the EU from the sanctioning economies, and any sanction case enforced by at least one EU country, in the GSDB drops 506 of 1794 cases. The y-axis is normalized to the 2015 levels and should not be read as a count level.

Figure 22: Counterfactual Shift Share of Success Rate of Sanction Cases by Type Across Time, Excluding a Major Contemporaneous Target



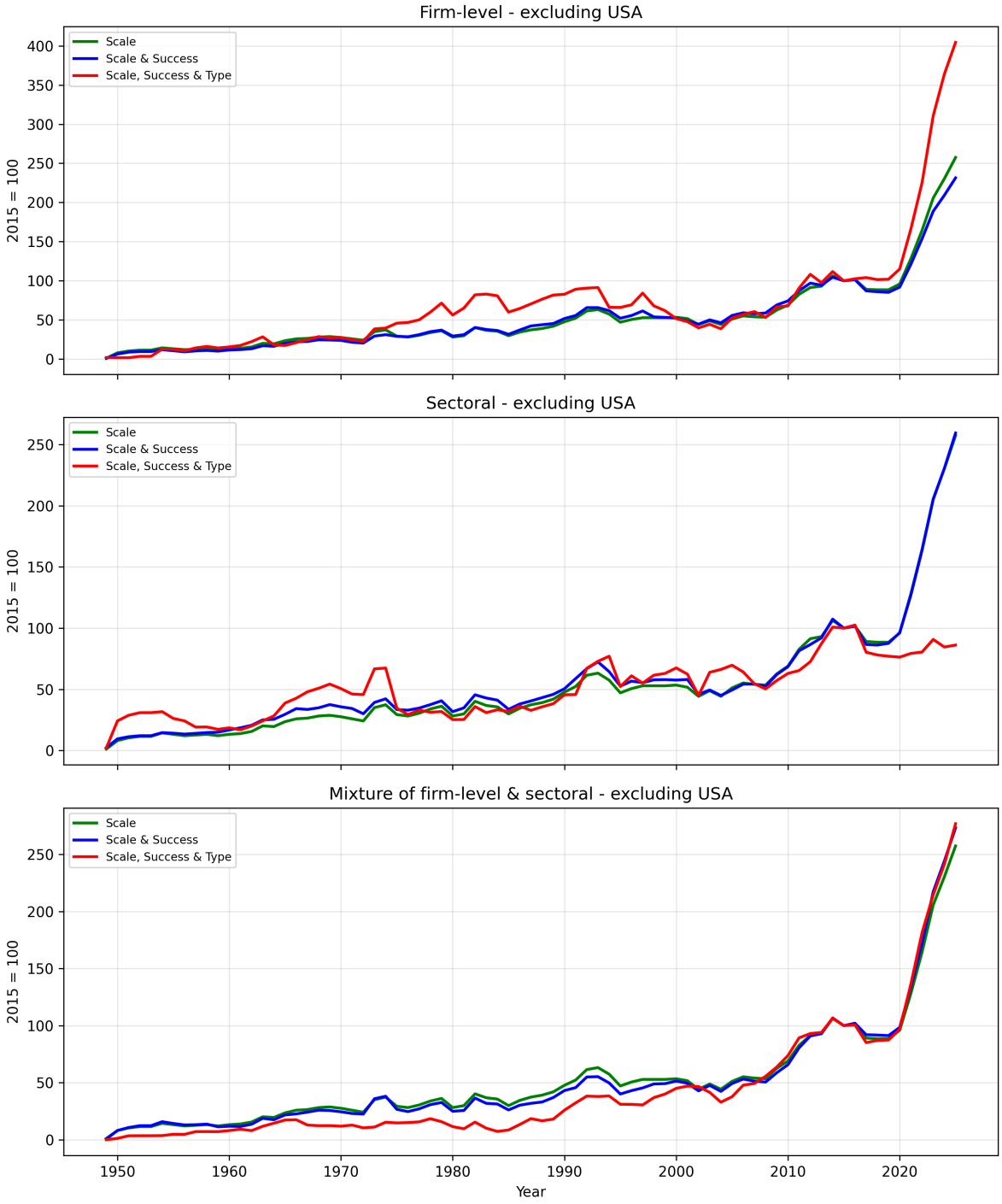
Notes: : This figure plots observed and counterfactual trends in evolutions of total number of active sanctions by type based on the statistical decomposition from equation (1). The green line plots the counterfactual level of sanctions by type with the same composition of success rate and relative use of sanction types inside every success rate in 2015. The blue line represents the level of sanctions by type, with the same use of sanction types inside every success rate as in 2015. The red line represents the actual observed number of sanctions by type, which consists of changes to the scale, composition of success rate, and type of sanctions relative to 2015 values. The top panel does the decomposition for sanction cases solely using firm-level sanctions. The middle panel does the same decomposition for sanctions using solely sectoral sanctions. The bottom panel studies cases of sanction mixing firm-level and sectoral sanctions.
Excluding Russia as a target from the GSDB drops 75 of 1794 cases. The y-axis is normalized to the 2015 levels and should not be read as a count level.

Figure 23: Counterfactual Shift Share of Success Rate of Sanction Cases by Type Across Time, Excluding a Major Contemporaneous Target



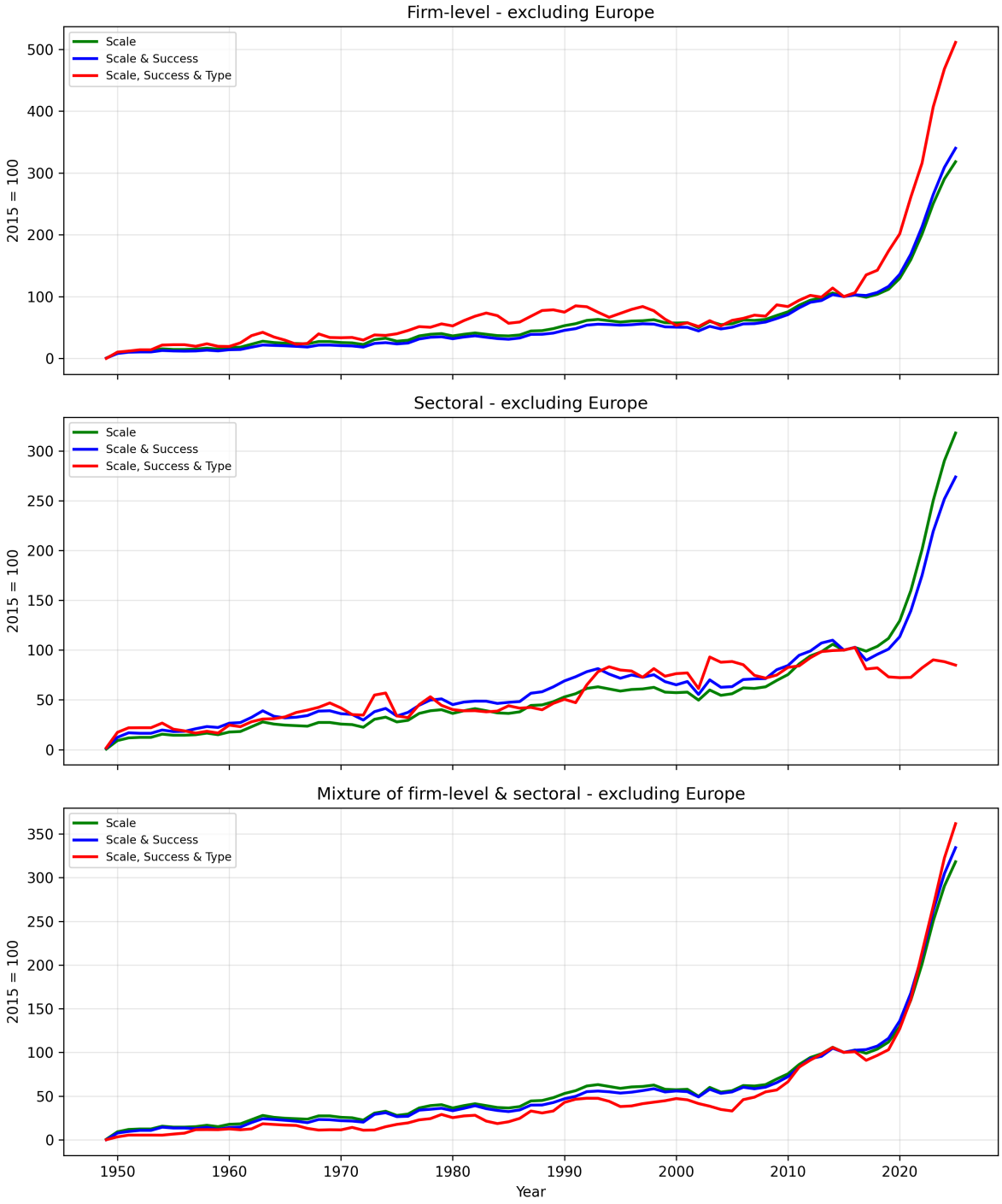
Notes: : This figure plots observed and counterfactual trends in evolutions of total number of active sanctions by type based on the statistical decomposition from equation (1). The green line plots the counterfactual level of sanctions by type with the same composition of success rate and relative use of sanction types inside every success rate in 2015. The blue line represents the level of sanctions by type, with the same use of sanction types inside every success rate as in 2015. The red line represents the actual observed number of sanctions by type, which consists of changes to the scale, composition of success rate, and type of sanctions relative to 2015 values. The top panel does the decomposition for sanction cases solely using firm-level sanctions. The middle panel does the same decomposition for sanctions using solely sectoral sanctions. The bottom panel studies cases of sanction mixing firm-level and sectoral sanctions.
Excluding China as a target from the GSDB drops 57 of 1794 cases. The y-axis is normalized to the 2015 levels and should not be read as a count level.

Figure 24: Counterfactual Shift Share of Success Rate of Sanction Cases by Type Across Time, Excluding a Major Sanctioning Economy



Notes: : This figure plots observed and counterfactual trends in evolutions of total number of active sanctions by type based on the statistical decomposition from equation (1). The green line plots the counterfactual level of sanctions by type with the same composition of success rate and relative use of sanction types inside every success rate in 2015. The blue line represents the level of sanctions by type, with the same use of sanction types inside every success rate as in 2015. The red line represents the actual observed number of sanctions by type, which consists of changes to the scale, composition of success rate, and type of sanctions relative to 2015 values. The top panel does the decomposition for sanction cases solely using firm-level sanctions. The middle panel does the same decomposition for sanctions using solely sectoral sanctions. The bottom panel studies cases of sanction mixing firm-level and sectoral sanctions.
Excluding the USA from the sanctioning economy in the GSDB drops 686 of 1794 cases. The y-axis is normalized to the 2015 levels and should not be read as a count level.

Figure 25: Counterfactual Shift Share of Success Rate of Sanction Cases by Type Across Time, Excluding a Major Sanctioning Economy



Notes: : This figure plots observed and counterfactual trends in evolutions of total number of active sanctions by type based on the statistical decomposition from equation (1). The green line plots the counterfactual level of sanctions by type with the same composition of success rate and relative use of sanction types inside every success rate in 2015. The blue line represents the level of sanctions by type, with the same use of sanction types inside every success rate as in 2015. The red line represents the actual observed number of sanctions by type, which consists of changes to the scale, composition of success rate, and type of sanctions relative to 2015 values. The top panel does the decomposition for sanction cases solely using firm-level sanctions. The middle panel does the same decomposition for sanctions using solely sectoral sanctions. The bottom panel studies cases of sanction mixing firm-level and sectoral sanctions.
Excluding the EU from the sanctioning economies, and any sanction case enforced by at least one EU country, in the GSDB drops 506 of 1794 cases. The y-axis is normalized to the 2015 levels and should not be read as a count level.